Top 54 CAT 2D & 3D LR Questions With Video Solutions

Questions

Instructions

Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from M to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute.
The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.

Question 1

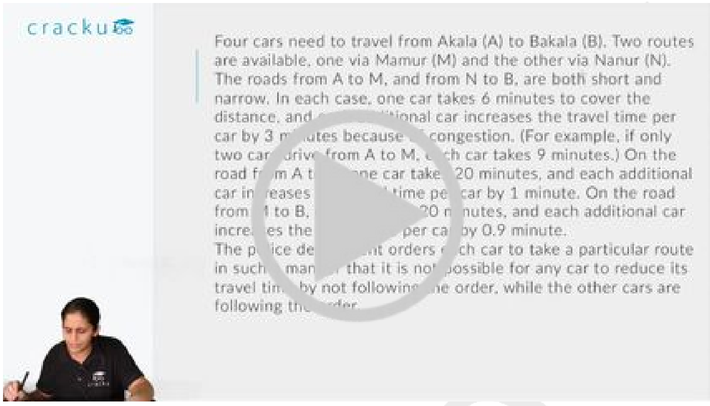
If all the cars follow the police order, what is the difference in travel time (in minutes) between a car which takes the route A-N-B and a car that takes the route A-M-B?

A 1

B 0.1

C 0.2

D 0.9



VIDEO SOLUTION

Question 2

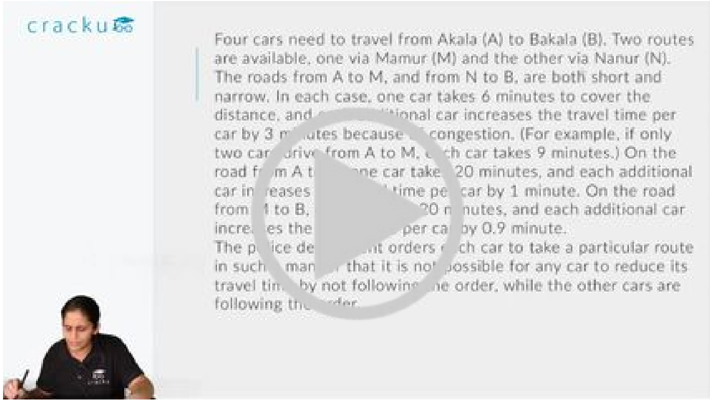
A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by j. minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.
If all the cars follow the police order, what is the minimum travel time (in minutes) from A to B? (Assume that the police department would never order all the cars to take the same route.)

A 26

B 32

C 29.9

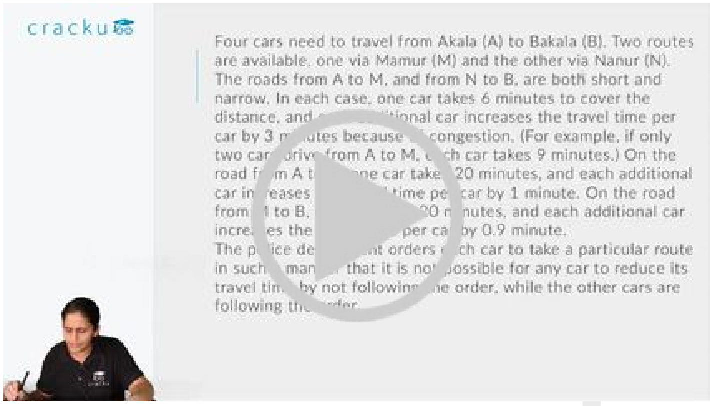
D 30



VIDEO SOLUTION

Question 3

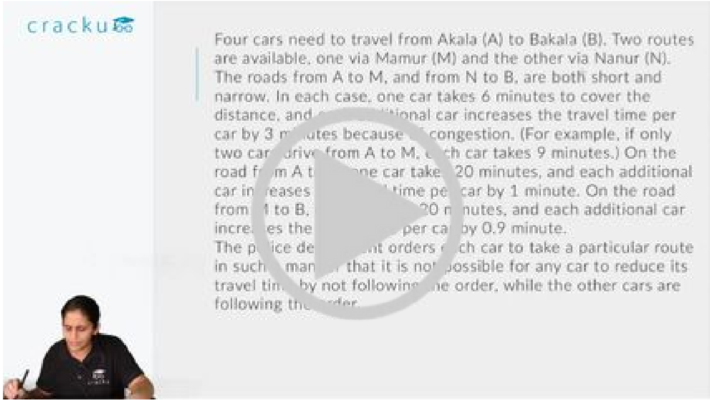
How many cars would be asked to take the route A-N-B, that is Akala-Nanur-Bakala route, by the police department?



VIDEO SOLUTION

Question 4

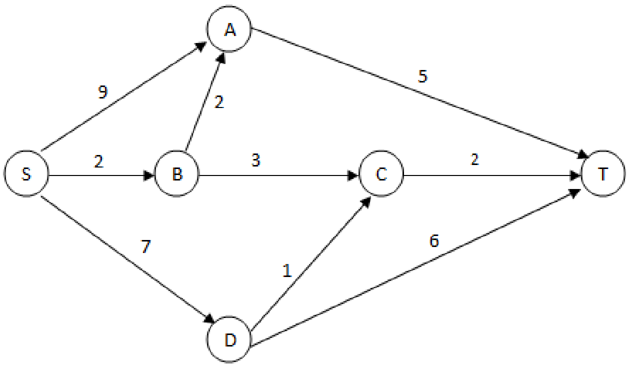
A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.
How many cars would the police department order to take the A-M-N-B route so that it is not possible for any car to reduce its travel time by not following the order while the other cars follow the order? (Assume that the police department would never order all the cars to take the same route.)



VIDEO SOLUTION

Instructions

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street. –



Motorists traveling from point S to point T would obviously take the route for which the total cost of traveling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

Question 5

If the government wants to ensure that no traffic flows on the street from D to T, while equal amount of traffic flows through junctions A and C, then a feasible set of toll charged (in rupees) at junctions A, B, C,and D respectively to achieve this goal is:

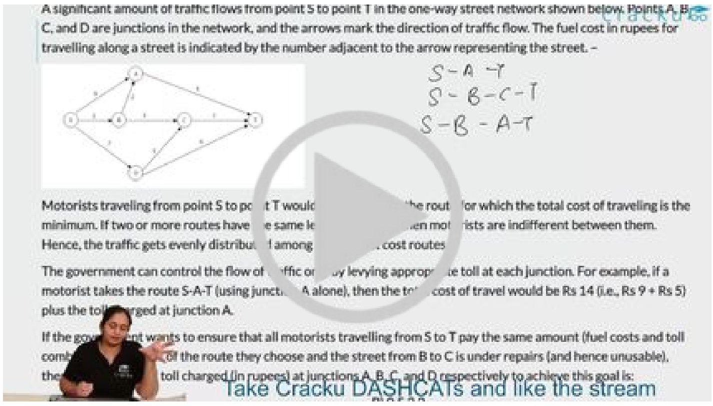
A 1,5,3,3

B 1,4,4,3

C 1,5,4,2

D 0,5,2,3

E 0,5,2,2



VIDEO SOLUTION

Question 6

If the government wants to ensure that all routes from S to T get the same amount of traffic, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

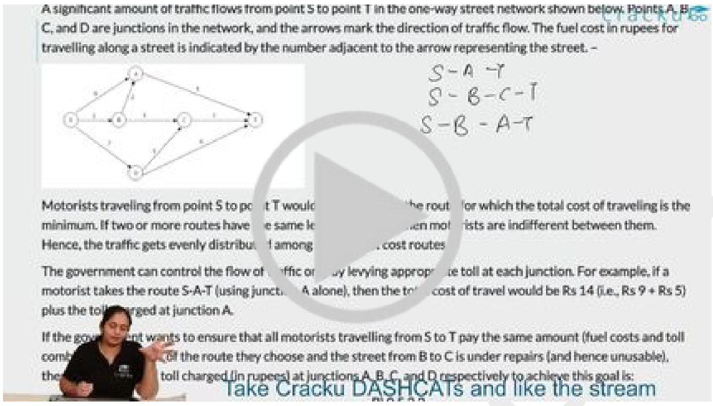
A 0, 5, 2, 2

B 0,5,4,1

C 1,5,3,3

D 1, 5, 3,2

E 1,5,4,2



VIDEO SOLUTION

Question 7

The government wants to devise a toll policy such that the total cost to the commuters per trip is minimized. The policy should also ensure that not more than 70 per cent of the total traffic passes through junction B. The cost incurred by the commuter travelling from point S to point T under this policy will be:

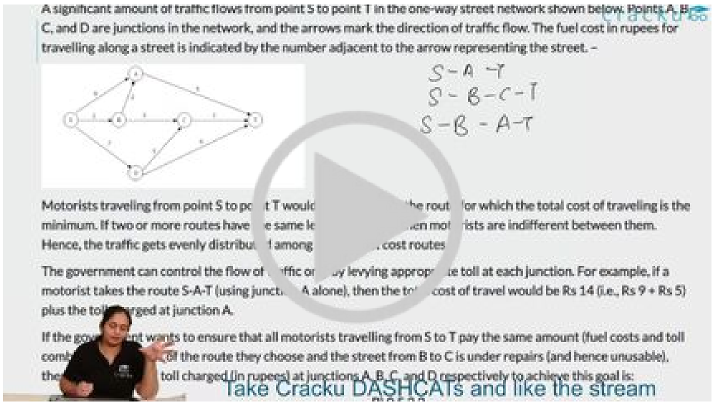
A Rs 7

B Rs 9

C Rs 10

D Rs 13

E Rs 14



VIDEO SOLUTION

Question 8

If the government wants to ensure that the traffic at S gets evenly distributed along streets from S to A, from S to B, and from S to D, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

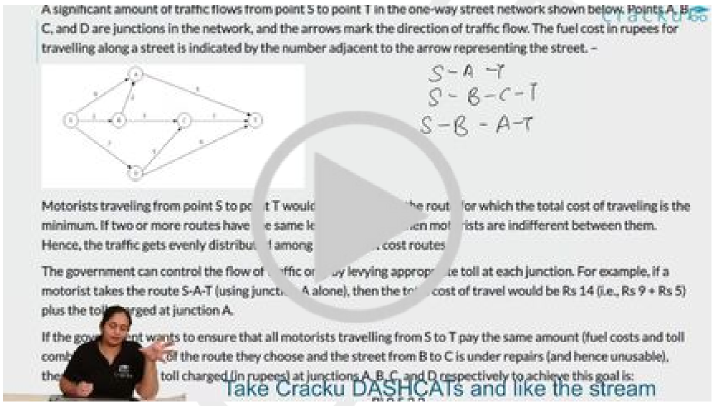
A 0,5,4,1

B 0,5,2,2

C 1,5,3,3

D 1,5,3,2

E 0,4,3,2



VIDEO SOLUTION

Question 9

If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

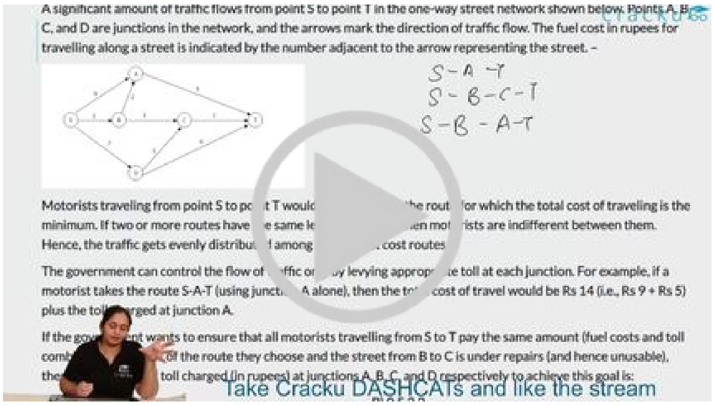
A 2,5,3,2

B 0,5,3,2

C 1,5,3,2

D 2,3,5,1

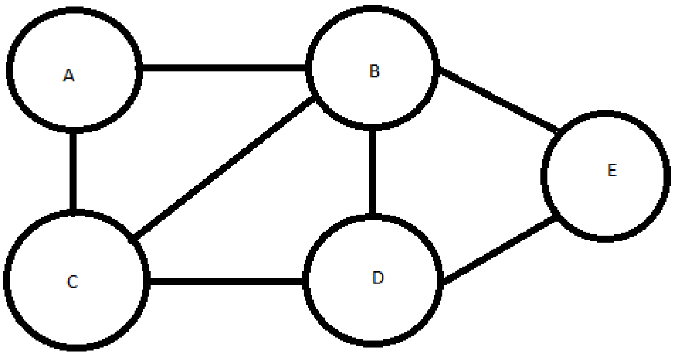
E 1,3,5,1



VIDEO SOLUTION

Instructions

There are 5 cities, A, B, C, D and E connected by 7 roads as shown in the figure below:



Design a route such that you start from any city of your choice and walk on each of the 7 roads once and only once, not necessarily returning to the city from which you started.

Question 10

For a route that satisfies the above restrictions, which of the following statements is true?

A There is no route that satisfies the above restriction.

B A route can either start at C or end at C, but not both.

C D can be only an intermediate city in the route.

D The route has to necessarily end at E.

VIEW SOLUTION

Question 11

How many different starting cities are possible such that the above restriction is satisfied?

A one

B zero

C three

D two

VIEW SOLUTION

Instructions

Direction for questions 108 and 109: Answer the questions based on the following information. In a locality, there are five small cities: A, B, C, D and E. The distances of these cities from each other are as follows. AB = 2 km AC = 2km AD > 2 km AE > 3 km BC = 2 km BD = 4 km BE = 3 km CD = 2 km CE = 3 km DE > 3 km

Question 12

If a ration shop is to be set up within 3 km of each city, how many ration shops will be required?

A 1

B 2

C 3

D 4

VIEW SOLUTION

Question 13

If a ration shop is to be set up within 2 km of each city, how many ration shops will be required?

A 2

B 3

C 4

D 5

VIEW SOLUTION

Instructions

A new airlines company is planning to start operations in a country. The company has identified ten different cities which they plan to connect through their network to start with. The flight duration between any pair of cities will be less than one hour. To start operations, the company has to decide on a daily schedule.
The underlying principle that they are working on is the following:
Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.

Question 14

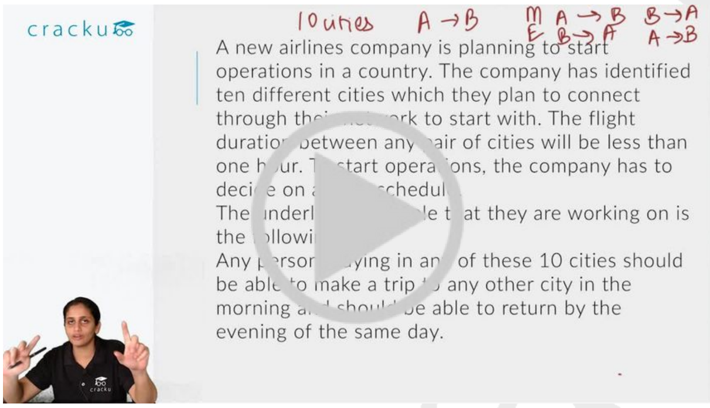
Suppose three of the ten cities are to be developed as hubs. A hub is a city which is connected with every other city by direct flights each way, both in the morning as well as in the evening. The only direct flights which will be scheduled are originating and/or terminating in one of the hubs. Then the minimum number of direct flights that need to be scheduled so that the underlying principle of the airline to serve all the ten cities is met without visiting more than one hub during one trip is:

A 54

B 120

C 96

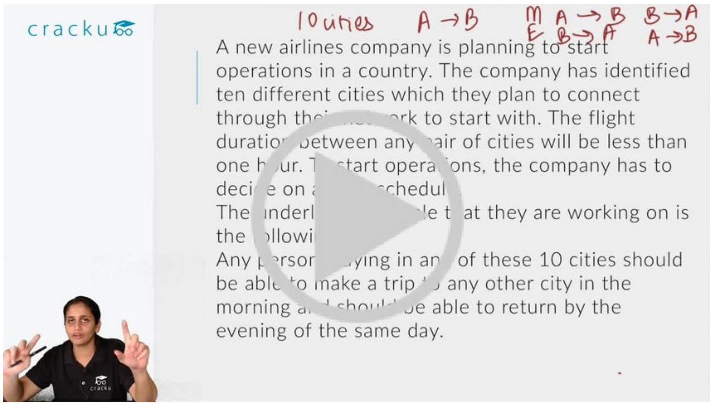
D 60



VIDEO SOLUTION

Question 15

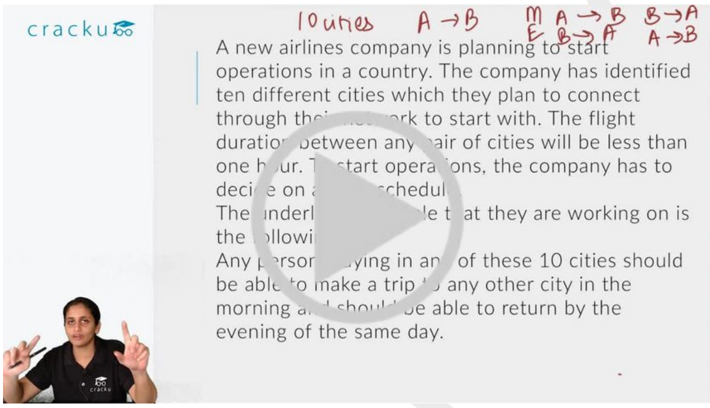
Suppose the 10 cities are divided into 4 distinct groups G1, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that G1 consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:
1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4
Then the minimum number of direct flights that satisfies the underlying principle of the airline is:



VIDEO SOLUTION

Question 16

Suppose the 10 cities are divided into 4 distinct groups Gl, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that Gl consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:
1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4
However, due to operational difficulties at A, it was later decided that the only flights that would operate at A would be those to and from B. Cities in G2 would have to be assigned to G3 or to G4.
What would be the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose?



VIDEO SOLUTION

Question 17

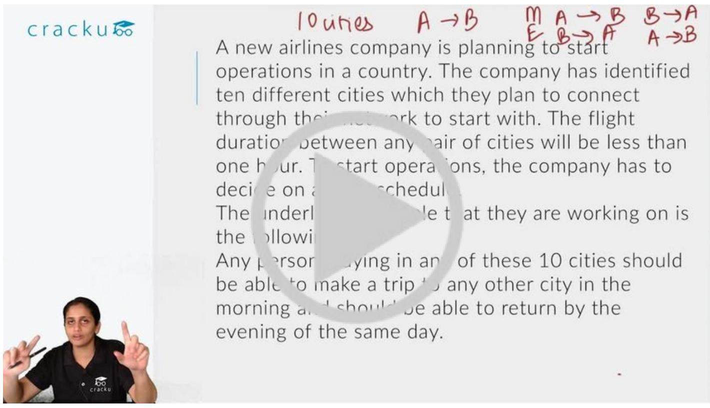
If the underlying principle is to be satisfied in such a way that the journey between any two cities can be performed using only direct (non-stop) flights, then the minimum number of direct flights to be scheduled is:

A 45

B 90

C 180

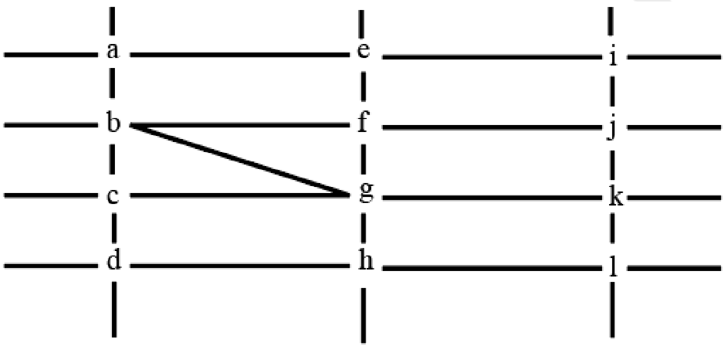
D 135



VIDEO SOLUTION

Instructions

The figure below shows the street map for a certain region with the street intersections marked from a through l. A person standing at an intersection can see along straight lines to other intersections that are in her line of sight and all other people standing at these intersections. For example, a person standing at intersection g can see all people standing at intersections b, c, e, f, h, and k. In particular, the person standing at intersection g can see the person standing at intersection e irrespective of whether there is a person standing at intersection f.



Six people U, V, W, X, Y, and Z, are standing at different intersections. No two people are standing at the same intersection.
The following additional facts are known.
1. X, U, and Z are standing at the three corners of a triangle formed by three street segments.
2. X can see only U and Z.
3. Y can see only U and W.
4. U sees V standing in the next intersection behind Z.
5. W cannot see V or Z.
6. No one among the six is standing at intersection d.

Question 18

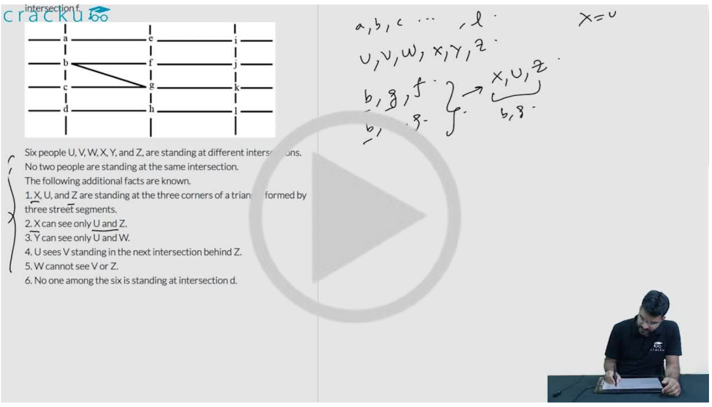
Who can V see?

A Z only

B U, W and Z only

C U and Z only

D U only



VIDEO SOLUTION

Question 19

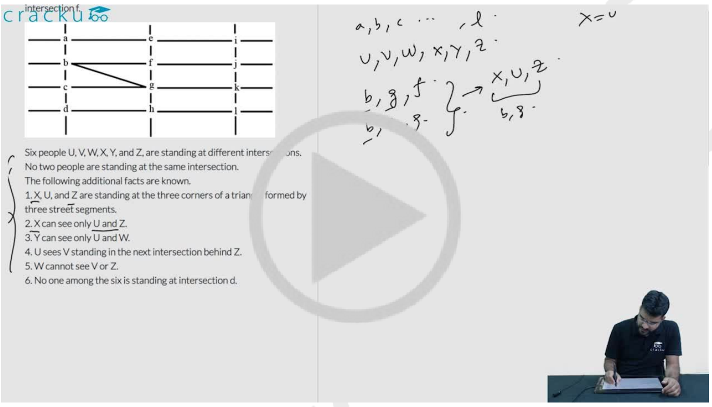
What is the minimum number of street segments that X must cross to reach Y?

A 1

B 4

C 2

D 3



VIDEO SOLUTION

Question 20

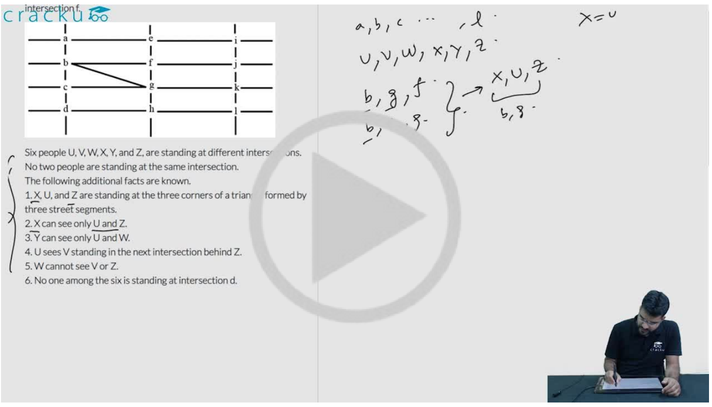
Should a new person stand at intersection d, who among the six would she see?

A W and X only

B U and W only

C U and Z only

D V and X only



VIDEO SOLUTION

Question 21

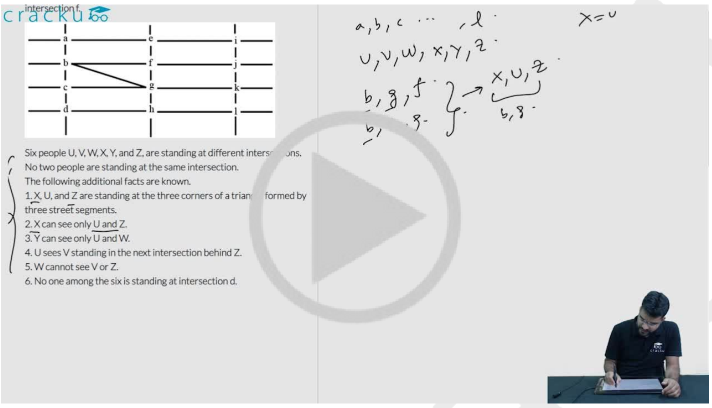
Who is standing at intersection a?

A W

B Y

C No one

D V



VIDEO SOLUTION

Instructions

In an 8 X 8 chess board a queen placed any where can attack another piece if the piece is present in the same row, or in the same column or in any diagonal position in any possible 4 directions, provided there is no other piece in between in the path from the queen to that piece.

The columns are labelled a to h (left to right) and the rows are numbered 1 to 8 (bottom to top). The position of a piece is given by the combination of column and row labels. For example, position c5 means that the piece is in c^{th} column and 5^{th} row.

Question 22

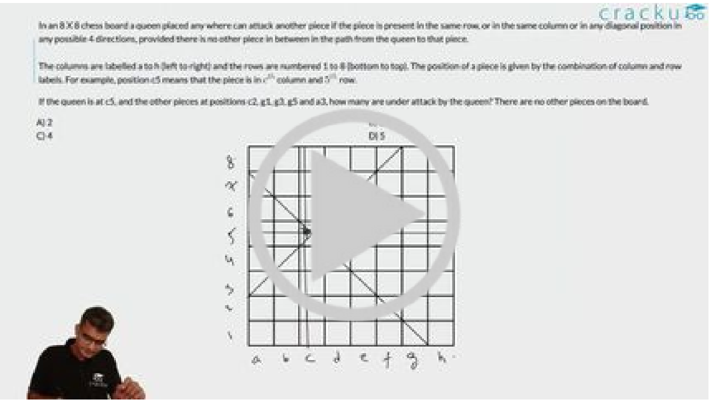
If the other pieces are only at positions a1, a3, b4, d7, h7 and h8, then which of the following positions of the queen results in the maximum number of pieces being under attack?

A f8

B a7

C c1

D d3



VIDEO SOLUTION

Question 23

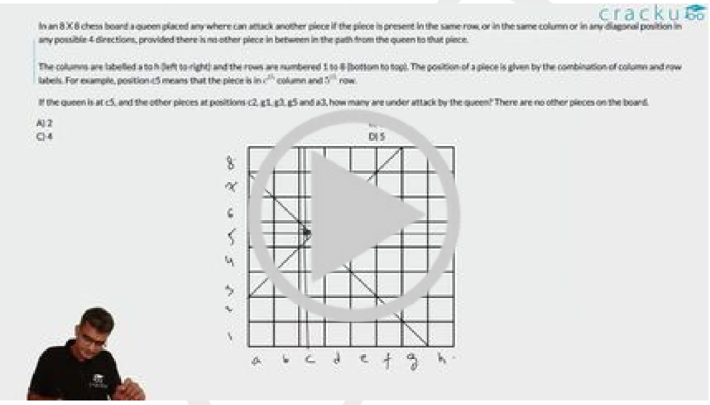
If the other pieces are only at positions a1, a3, b4, d7, h7 and h8, then from how many positions the queen cannot attack any of the pieces?

A 0

B 3

C 4

D 6



VIDEO SOLUTION

Question 24

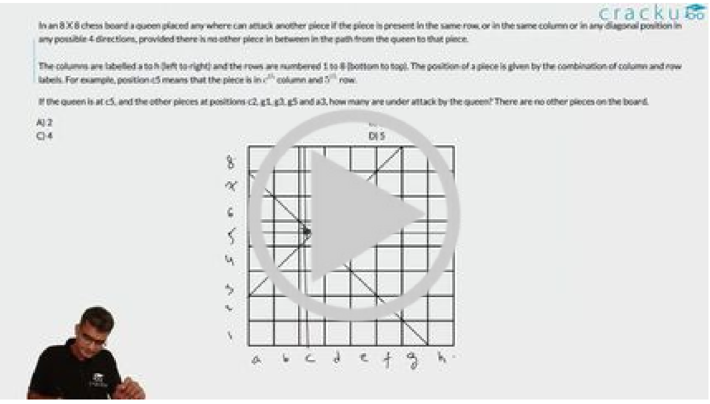
If the queen is at c5, and the other pieces at positions c2, g1, g3, g5 and a3, how many are under attack by the queen? There are no other pieces on the board.

A 2

B 3

C 4

D 5



VIDEO SOLUTION

Question 25

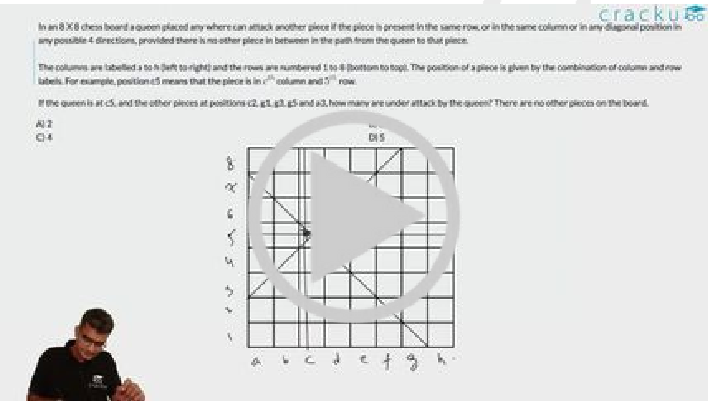
Suppose the queen is the only piece on the board and it is at position d5. In how many positions can another piece be placed on the board such that it is safe from attack from the queen?

A 32

B 35

C 36

D 37



VIDEO SOLUTION

Instructions

You are given an n×n square matrix to be filled with numerals so that no two adjacent cells have the same numeral. Two cells are called adjacent if they touch each other horizontally, vertically or diagonally. So a cell in one of the four corners has three cells adjacent to it, and a cell in the first or last row or column which is not in the corner has five cells adjacent to it. Any other cell has eight cells adjacent to it.

Question 26

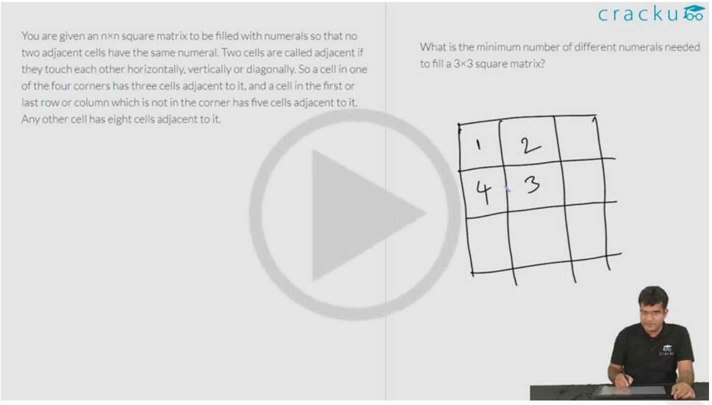
Suppose you are allowed to make one mistake, that is, one pair of adjacent cells can have the same numeral. What is the minimum number of different numerals required to fill a 5×5 matrix?

A 4

B 16

C 9

D 25



VIDEO SOLUTION

Question 27

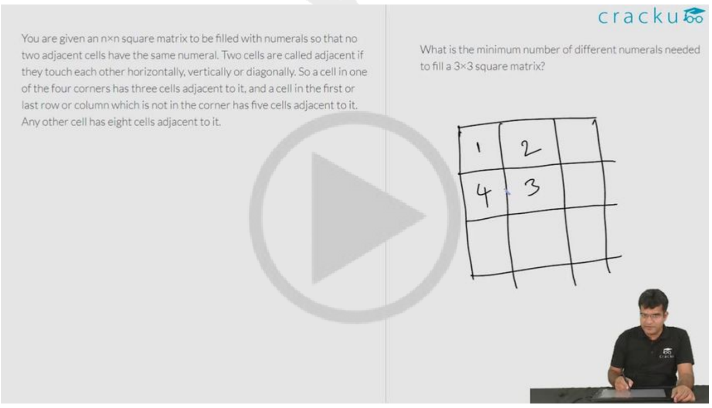
Suppose that all the cells adjacent to any particular cell must have different numerals. What is the minimum number of different numerals needed to fill a 5×5 square matrix?

A 25

B 4

C 16

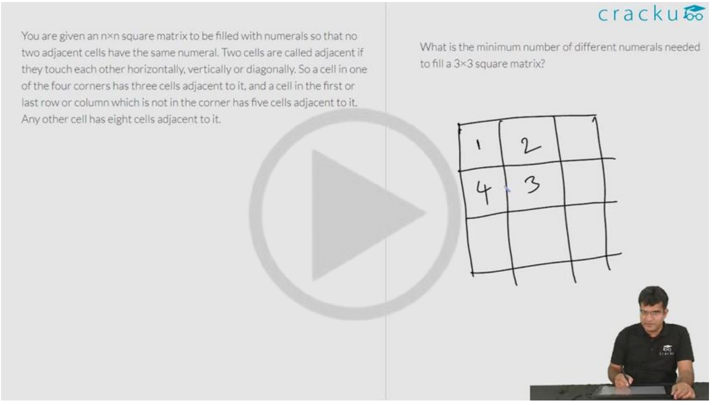
D 9



VIDEO SOLUTION

Question 28

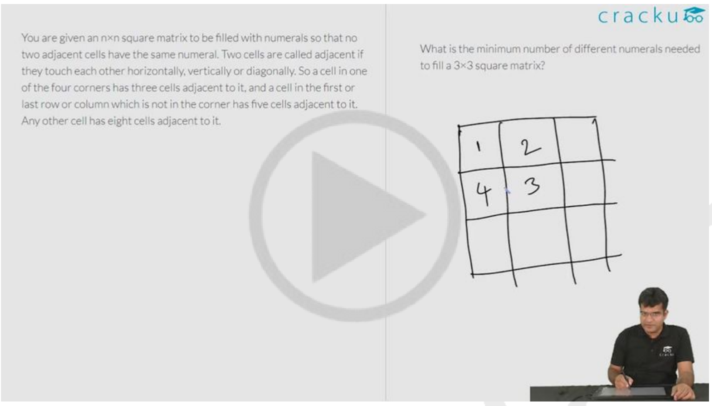
What is the minimum number of different numerals needed to fill a 5×5 square matrix?



VIDEO SOLUTION

Question 29

What is the minimum number of different numerals needed to fill a 3×3 square matrix?

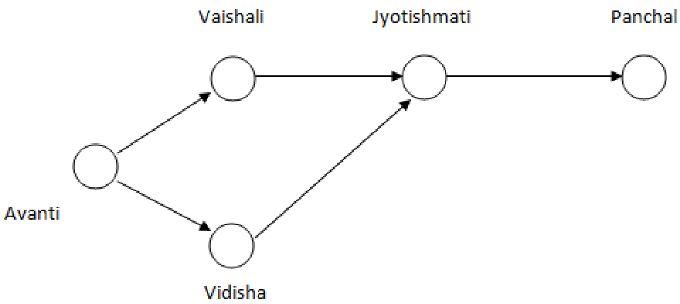


VIDEO SOLUTION

Instructions

Directions for the following three questions: Answer the questions based on the pipeline diagram below.

The following sketch shows the pipelines carrying material from one location to another. Each location has a demand for material. The demand at Vaishali is 400, at Jyotishmati is 400, at Panchal is 700, and at Vidisha is 200. Each arrow indicates the direction of material flow through the pipeline. The flow from Vaishali to Jyotishmati is 300. The quantity of material flow is such that the demands at all these locations are exactly met. The capacity of each pipeline is 1,000.



Question 30

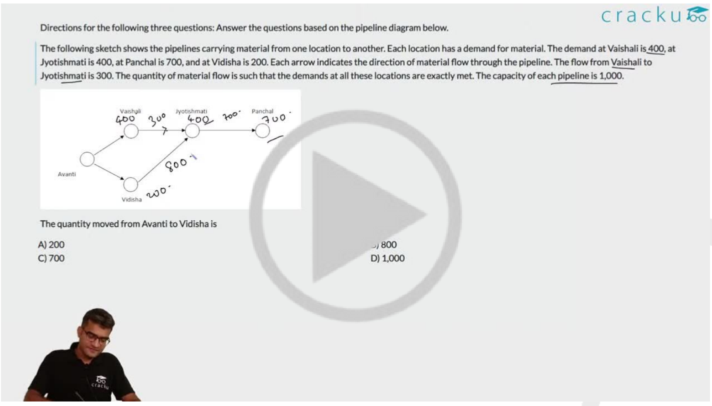
The free capacity available at the Avanti-Vaishali pipeline is

A 0

B 100

C 200

D 300



VIDEO SOLUTION

Question 31

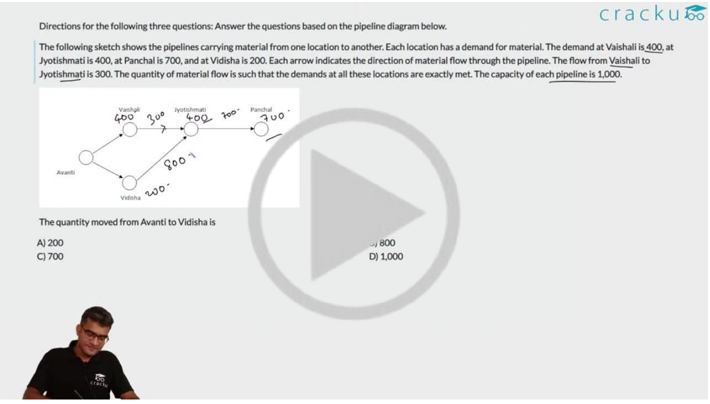
What is the free capacity available in the Avanti-Vidisha pipeline?

A 300

B 200

C 100

D 0



VIDEO SOLUTION

Question 32

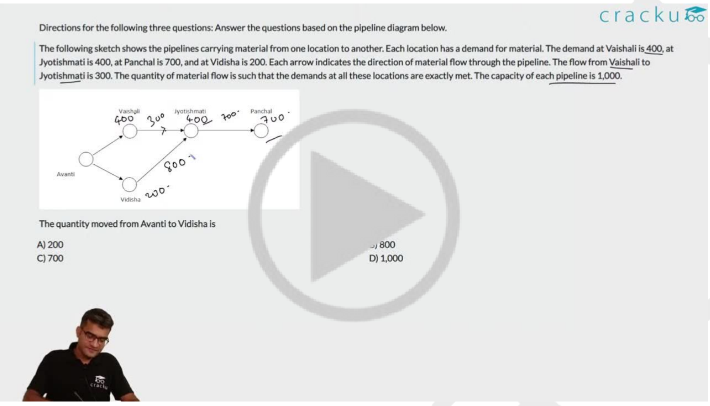
The quantity moved from Avanti to Vidisha is

A 200

B 800

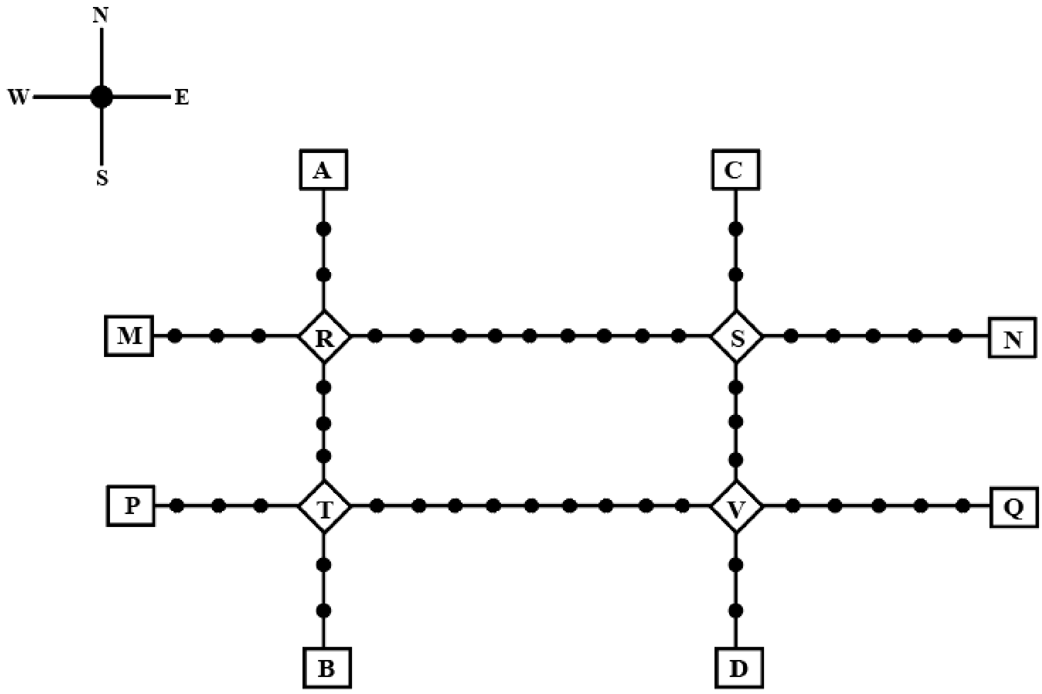
C 700

D 1,000



VIDEO SOLUTION

Instructions



Given above is the schematic map of the metro lines in a city with rectangles denoting terminal stations (e.g. A), diamonds denoting junction stations (e.g. R) and small filled-up circles denoting other stations. Each train runs either in east-west or north-south direction, but not both. All trains stop for 2 minutes at each of the junction stations on the way and for 1 minute at each of the other stations. It takes 2 minutes to reach the next station for trains going in east-west direction and 3 minutes to reach the next station for trains going in northsouth direction. From each terminal station, the first train starts at 6 am; the last trains leave the terminal stations at midnight. Otherwise, during the service hours, there are metro service every 15 minutes in the north-south lines and every 10 minutes in the east-west lines. A train must rest for at least 15 minutes after completing a trip at the terminal station, before it can undertake the next trip in the reverse direction. (All questions are related to this metro service only. Assume that if someone reaches a station exactly at the time a train is supposed to leave, (s)he can catch that train.)

Question 33

What is the minimum number of trains that are required to provide the service in this city?

VIDEO SOLUTION

Question 34

What is the minimum number of trains that are required to provide the service on the AB line (considering both north and south directions)?

VIDEO SOLUTION

Question 35

If Hari is ready to board a train at 8:05 am from station M, then when is the earliest that he can reach station N?

A 9:11 am

B 9:06 am

C 9:01 am

D 9:13 am

VIDEO SOLUTION

Question 36

If Priya is ready to board a train at 10:25 am from station T, then when is the earliest that she can reach station S?

A 11:12 am

B 11:22 am

C 11:07 am

D 11:28 am

VIDEO SOLUTION

Question 37

Haripriya is expected to reach station S late. What is the latest time by which she must be ready to board at station S if she must reach station B before 1 am via station R?

A 11:39 pm

B 11:49 am

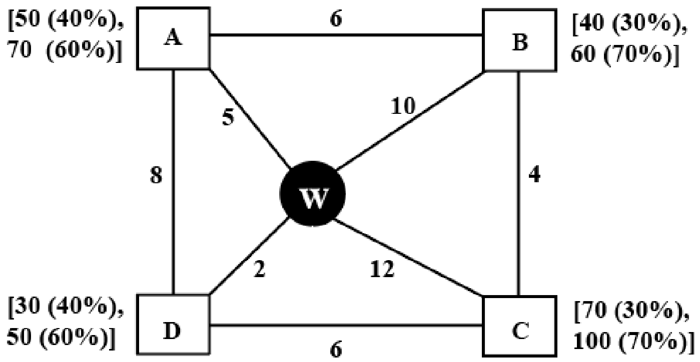
C 11:35 pm

D 11:43 pm

VIDEO SOLUTION

Instructions

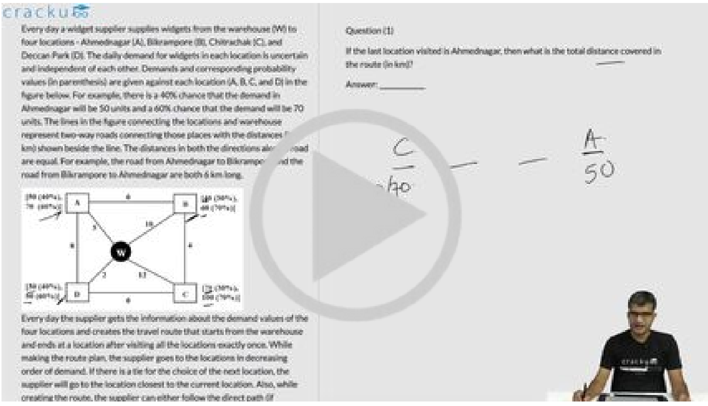
Every day a widget supplier supplies widgets from the warehouse (W) to four locations - Ahmednagar (A), Bikrampore (B), Chitrachak (C), and Deccan Park (D). The daily demand for widgets in each location is uncertain and independent of each other. Demands and corresponding probability values (in parenthesis) are given against each location (A, B, C, and D) in the figure below. For example, there is a 40% chance that the demand in Ahmednagar will be 50 units and a 60% chance that the demand will be 70 units. The lines in the figure connecting the locations and warehouse represent two-way roads connecting those places with the distances (in km) shown beside the line. The distances in both the directions along a road are equal. For example, the road from Ahmednagar to Bikrampore and the road from Bikrampore to Ahmednagar are both 6 km long.



Every day the supplier gets the information about the demand values of the four locations and creates the travel route that starts from the warehouse and ends at a location after visiting all the locations exactly once. While making the route plan, the supplier goes to the locations in decreasing order of demand. If there is a tie for the choice of the next location, the supplier will go to the location closest to the current location. Also, while creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse. If both paths are available (direct and via warehouse), the supplier will choose the path with minimum distance.

Question 38

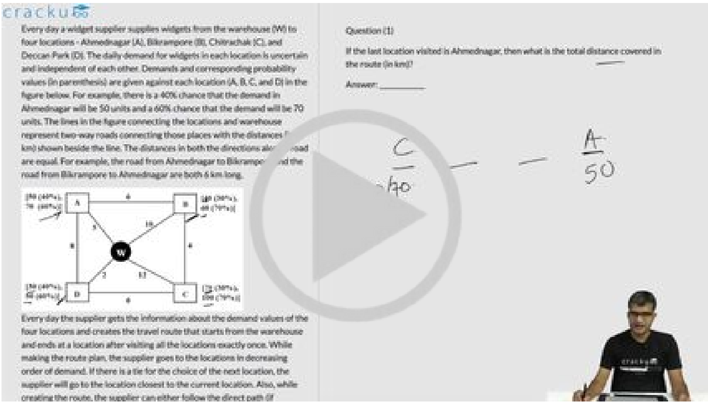
If the last location visited is Ahmednagar, then what is the total distance covered in the route (in km)?



VIDEO SOLUTION

Question 39

If the total number of widgets delivered in a day is 250 units, then what is the total distance covered in the route (in km)?



VIDEO SOLUTION

Question 40

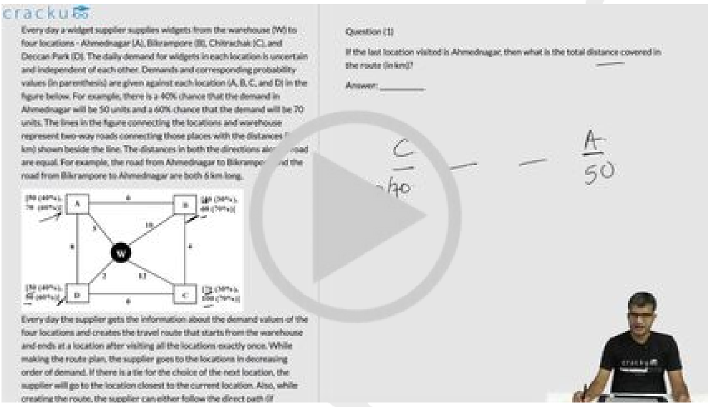
If the first location visited from the warehouse is Ahmednagar, then what is the chance that the total distance covered in the route is 40 km?

A 18%

B 5.4%

C 3.24%

D 30%



VIDEO SOLUTION

Question 41

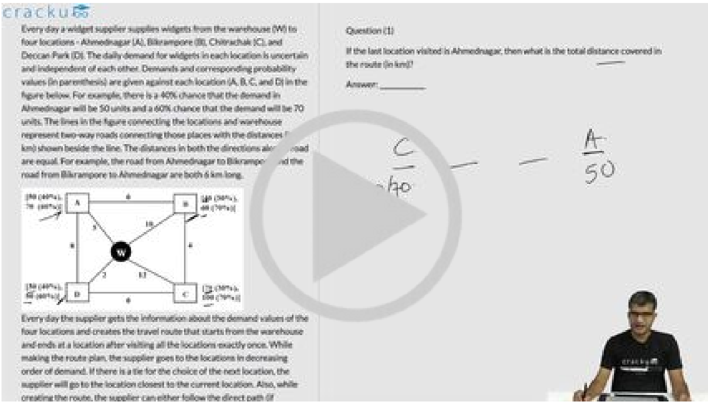
If Ahmednagar is not the first location to be visited in a route and the total route distance is 29 km, then which of the following is a possible number of widgets delivered on that day?

A 210

B 220

C 200

D 250



VIDEO SOLUTION

Question 42

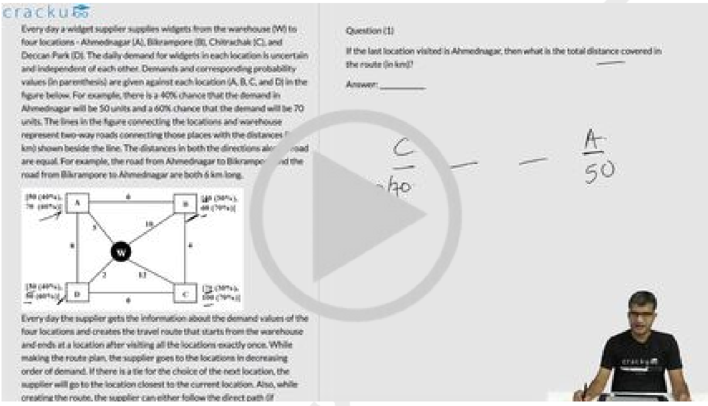
What is the chance that the total number of widgets delivered in a day is 260 units and the route ends at Bikrampore?

A 33.33%

B 10.80%

C 17.64%

D 7.56%



VIDEO SOLUTION

Instructions

A farmer had a rectangular land containing 205 trees. He distributed that land among his four daughters - Abha, Bina, Chitra and Dipti by dividing the land into twelve plots along three rows (X,Y,Z) and four Columns (1,2,3,4) as shown in the figure below:

	1	2	3	4
X	12 C			
Y	21 A			A
Z	B	C	9	28

The plots in rows X, Y, Z contained mango, teak and pine trees respectively. Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees. Each daughter got an even number of plots. In the figure, the number mentioned in top left corner of a plot is the number of trees in that plot, while the letter in the bottom right corner is the first letter of the name of the daughter who got that plot (For example, Abha got the plot in row Y and column 1 containing 21 trees). Some information in the figure got erased, but the following is known:

1. Abha got 20 trees more than Chitra but 6 trees less than Dipti.
2. The largest number of trees in a plot was 32, but it was not with Abha.
3. The number of teak trees in Column 3 was double of that in Column 2 but was half of that in Column 4.
4. Both Abha and Bina got a higher number of plots than Dipti.
5. Only Bina, Chitra and Dipti got corner plots.
6. Dipti got two adjoining plots in the same row.
7. Bina was the only one who got a plot in each row and each column.
8. Chitra and Dipti did not get plots which were adjacent to each other (either in row / column / diagonal).
9. The number of mango trees was double the number of teak trees.

Question 43

Which column had the highest number of trees?

A 4

B 3

C Cannot be determined

D 2

VIDEO SOLUTION

Question 44

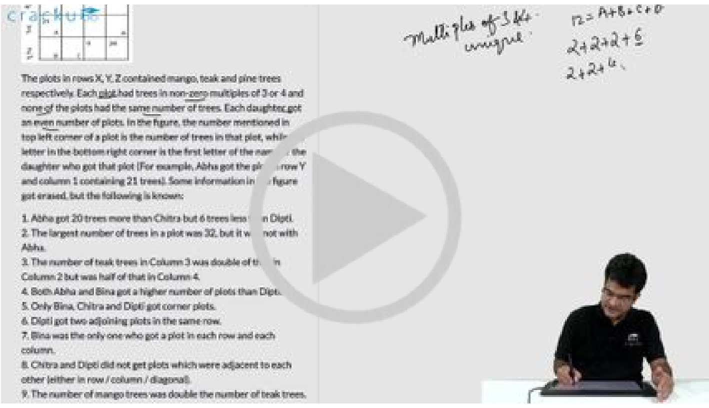
Who got the plot with the smallest number of trees and how many trees did that plot have?

A Dipti, 6 trees

B Bina, 3 trees

C Bina, 4 trees

D Abha, 4 trees



VIDEO SOLUTION

Question 45

How many pine trees did Chitra receive?

A 18

B 30

C 21

D 15

VIDEO SOLUTION

Question 46

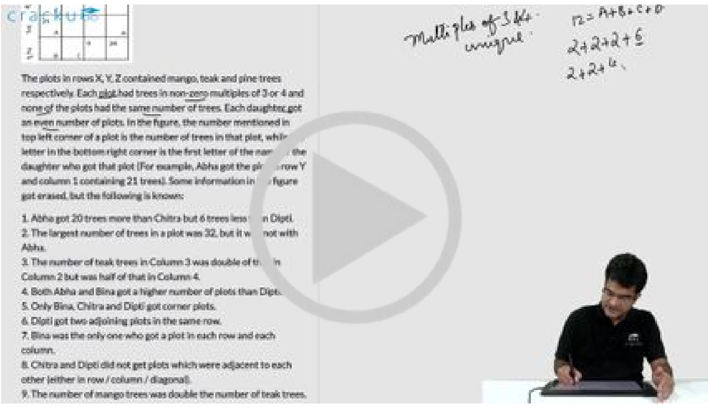
Which of the following is the correct sequence of trees received by Abha, Bina, Chitra and Dipti in that order?

A 50, 69, 30, 56

B 54, 57, 34, 60

C 44, 87, 24, 50

D 60, 39, 40, 66



VIDEO SOLUTION

Question 47

Which of the following statements is NOT true?

A Chitra got 12 mango trees

B Bina got 32 pine trees.

C Abha got 41 teak trees.

D Dipti got 56 mango trees

VIDEO SOLUTION

Question 48

How many mango trees were there in total?

A 49

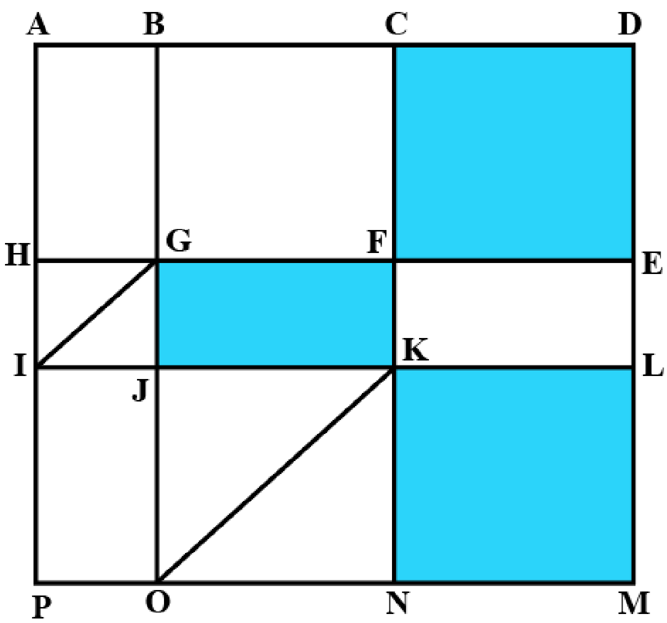
B 84

C 98

D 126

VIDEO SOLUTION

Instructions



The above is a schematic diagram of walkways (indicated by all the straight-lines) and lakes (3 of them, each in the shape of rectangles - shaded in the diagram) of a gated area. Different points on the walkway are indicated by letters (A through P) with distances being OP = 150 m, ON = MN = 300 m, ML = 400 m, EL = 200 m, DE = 400 m.

The following additional information about the facilities in the area is known.
1. The only entry/exit point is at C.
2. There are many residences within the gated area; all of them are located on the path AH and ML with four of them being at A, H, M, and L.
3. The post office is located at P and the bank is located at B.

Question 49

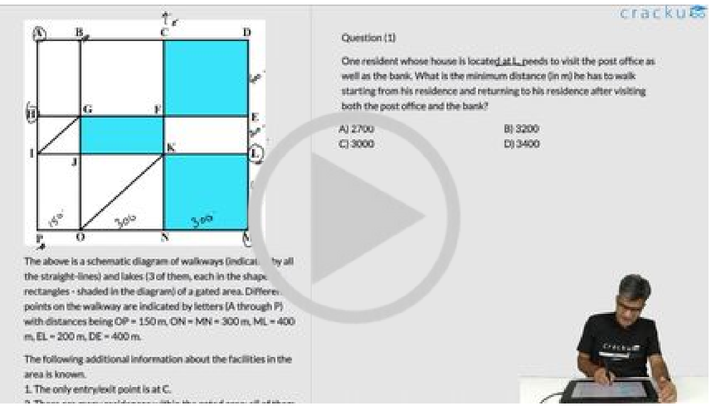
One resident whose house is located at L, needs to visit the post office as well as the bank. What is the minimum distance (in m) he has to walk starting from his residence and returning to his residence after visiting both the post office and the bank?

A 2700

B 3200

C 3000

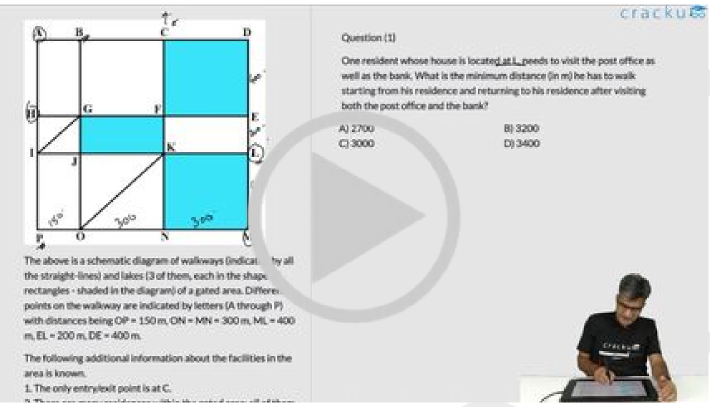
D 3400



VIDEO SOLUTION

Question 50

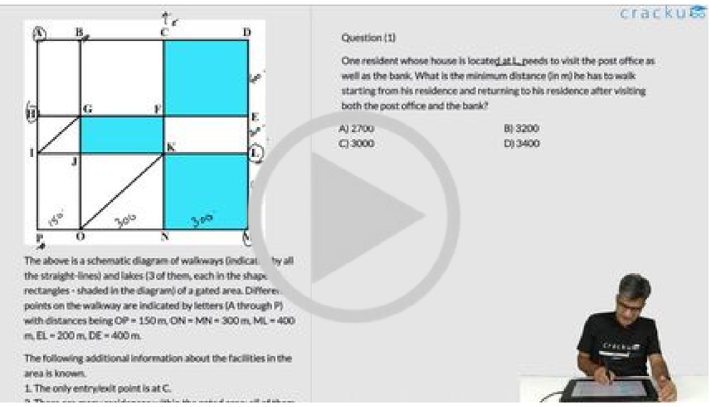
One resident takes a walk within the gated area starting from A and returning to A without going through any point (other than A) more than once. What is the maximum distance (in m) she can walk in this way?



VIDEO SOLUTION

Question 51

Visitors coming for morning walks are allowed to enter as long as they do not pass by any of the residences and do not cross any point (except C) more than once. What is the maximum distance (in m) that such a visitor can walk within the gated area?



VIDEO SOLUTION

Question 52

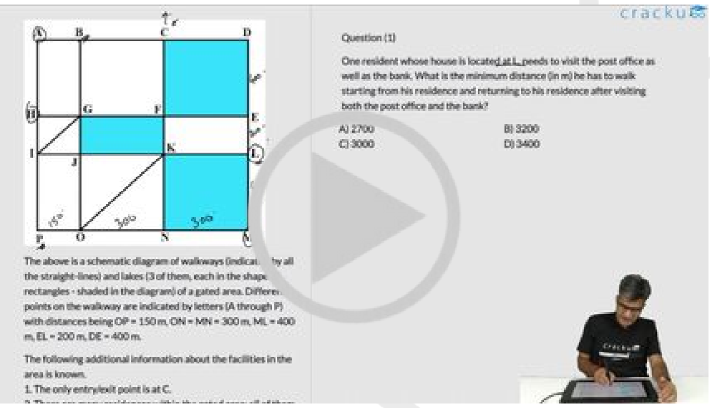
One person enters the gated area and decides to walk as much as possible before leaving the area without walking along any path more than once and always walking next to one of the lakes. Note that he may cross a point multiple times. How much distance (in m) will he walk within the gated area?

A 2800

B 3000

C 3800

D 3200



VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 53

In a six-node network, two nodes are connected to all the other nodes. Of the remaining four, each is connected to four nodes. What is the total number of links in the network?

A 13

B 15

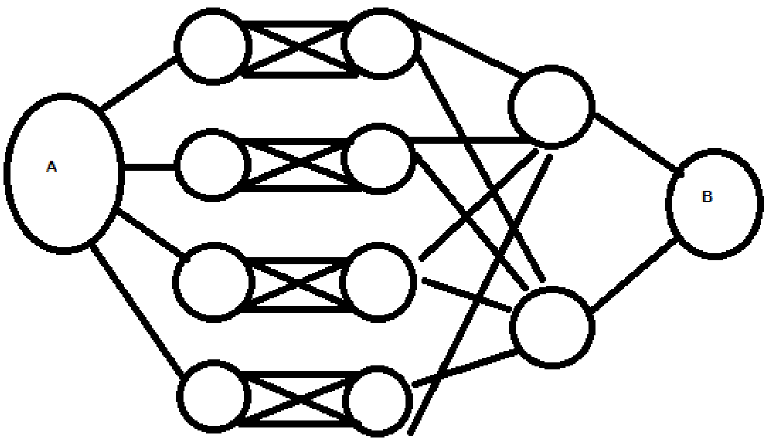
C 7

D 26

VIEW SOLUTION

Question 54

What is the total number of ways to reach A to B in the network given, such that no node is included twice and one can only move from left to right?



A 24

B 32

C 48

D 60

VIEW SOLUTION

Answers

1.**B**	2.**B**	3.**2**	4.**2**	5.**E**	6.**D**	7.**C**	8.**A**
9.**C**	10.**B**	11.**D**	12.**A**	13.**A**	14.**C**	15.**40**	16.**4**
17.**C**	18.**C**	19.**C**	20.**A**	21.**C**	22.**D**	23.**C**	24.**C**
25.**C**	26.**A**	27.**D**	28.**4**	29.**4**	30.**D**	31.**D**	32.**D**
33.**48**	34.**8**	35.**A**	36.**A**	37.**A**	38.**35**	39.**38**	40.**A**
41.**A**	42.**D**	43.**A**	44.**B**	45.**A**	46.**A**	47.**B**	48.**C**
49.**B**	50.**5100**	51.**3500**	52.**C**	53.**A**	54.**B**		

Explanations

1.**B**

Since, two cars are allowed on each route, The A-M part and N-B will take same time. The difference will be in travelling M-B part and A-N part, and that difference is 0.1 minute. In route M-B car will take 20+0.9 = 20.9 min and in route A-N car will take 20+1 = 21 min. So difference = 0.1 min

VIDEO SOLUTION

2.**B**

From the previous question we have found that

1 car take AMB route, 2 cars take AMNB route and other take ANB route.

Then the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then travel time of AMB will be A-M + M-B = (6+3*2) + (20) = 32

Then travel time of AMNB will be A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32

Then travel time of ANB will be A-N + N-B = (20) + (6+3*2) = 32

The minimum travel time from A to B is 32 min.

VIDEO SOLUTION

3.**2**

Since there are two routes i.e A-M-B and A-N-B and four cars, then 2 cars must be allowed to take each route. In case if one car tried to break rule, then its travel time will increase. Now assume that on route A-M-B three cars are allowed and on route A-N-B one car is allowed, then one car running on A-M-B can break the rule and reduce its travel time. Hence, two cars must be allowed on each route.

VIDEO SOLUTION

4.**2**

Case 1: Let us assume 1 car takes AMB route, 3 cars take ANB route
Then travel time of AMB will be A-M + M-B = 6 + 20 = 26
Then travel time of ANB will be A-N + N-B = (20+2) + (6+3*2) = 34

Now, one car(A) travelling on ANB broke the rule and decided to move on AMB route. then

Case 2: Let us assume 2 cars take AMB route, 2 cars take ANB route
Then travel time of AMB will be A-M + M-B =(6+3) + (20+0.9) = 29.9
Then travel time of ANB will be A-N + N-B = (20+1) + (6+3) = 30

Since the car A reduced its time from 34 to 29.9, the case-1 route is not optimal, hence case 1 is invalid.

Now, one car(B) travelling on ANB broke the rule and decided to move on AMB route. then, case

Case 3: Let us assume 3 cars take AMB route, 1 car take ANB route

Then travel time of AMB will be A-M + M-B = (6+3*2) + (20+0.9*2) = 33.8
Then travel time of ANB will be A-N + N-B = (20) + (6) = 29

Now, one car(C) travelling on AMB broke the rule and decided to move on AMNB route. then
Case 4: Let us assume 2 cars take AMB route, 1 car take AMNB route and other take ANB route.

Then travel time of AMB will be A-M + M-B = (6+3*2) + (20+0.9) = 32.9
Then travel time of AMNB will be A-M + M-N + N-B = (6+3*2) + (7) + (6+3) = 28
Then travel time of ANB will be A-N + N-B = (20) + (6+3) = 29

Since this car(C) reduced its time from 33.8 to 28, the route is not optimal, hence case 3 is invalid.
Since this car(B) reduced its time from 30 to 28, the route is not optimal, hence case 2 is invalid.
Now, one car(D) travelling on AMB broke the rule and decided to move on AMNB route. then

Case 5: Let us assume 1 car takes AMB route, 2 cars take AMNB route and other takes ANB route.

Then, the portion A-M will be travelled by 3 cars, M-B by one car, M-N by 2 cars, A-N by 1 car and N-B by 3 cars.

Then, travel time of AMB will be A-M + M-B = (6+3*2) + (20) = 32
Then, travel time of AMNB will be A-M + M-N + N-B = (6+3*2) + (7+1) + (6+3*2) = 32
Then, travel time of ANB will be A-N + N-B = (20) + (6+3*2) = 32

It is clear that the car D reduced its time from 32.9 min to 32 min if it broke the rule. Hence, case 4 is invalid.
In this arrangement of case 5, no car can improve their travel time by changing their path
Hence, the optimal allocation will be to order 2 cars on A-M-N-B route.

VIDEO SOLUTION

5. **E**

Let the toll charged at junctions A, B, C, and D be a,b,c and d respectively. Now since we want equal traffic through A and C , total cost through routes passing from A and C should be equal. So we have (9+a+5) + (2+b+2+a+5) = (2+3+b+c+2) + (7+d+1+c+2) . Only option E satisfy the above equality.

VIDEO SOLUTION

6. **D**

Now the fuel cost along different routes are :
SAT = 14
SBAT = 9
SBCT = 7
SDCT = 10
SDT = 13
Now , if we consider option D . Total cost for all routes comes out to be same which is 15. Hence option D.

VIDEO SOLUTION

7. **C**

The costs of the routes are as given below:
S - B - C - T = 7
S - B - A - T = 9
S - D - C - T = 10

S - D - T = 13
S - A -T = 14
Hence now 100% of the traffic flows through S - B - C - T
Now if we make the cost of traveling through S - B - C - T same as some other route not going through B, then the traffic will be equally distributed between these two routes. The lowest such route is S-D-C-T. The difference in cost = 3. Hence if we levy a toll of Rs.3 at B, the costs of SBCT and SBAT become 10,12 respectively and other routes are not affected. So 50% traffic flows through SBCT and 50% flows through SDCT . Hence cost in this policy = 10.

VIDEO SOLUTION

8. **A**
Total cost = fuel cost + toll
Total cost along SAT : 14+tollA
Total cost along SBAT : 9+tollA+tollB
Total cost along SDT : 13+tollD
Now when option A is considered , total costs come out to be same.
Hence option A is correct.

VIDEO SOLUTION

9. **C**
Let the toll charged at junctions A, B, C, and D be a,b,c and d respectively. Then the so that equal amount is collected through all route we have, 9+a+5=2+b+2+a+5=10+d+c=13+d. Then from the options only option C satisfies the above equality. hence option C.

VIDEO SOLUTION

10. **B**
We can start from either city C or D.

C - A - B - D - C - B - E - D

D - B - A - C - B - E - D - C

We cannot start from any other city.

So, we can either start from C or end at C but not both.

VIEW SOLUTION

11. **D**
We can start from either city C or D.

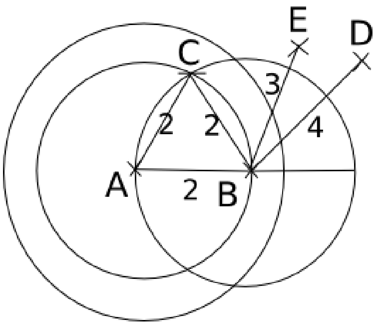
C - A - B - D - C - B - E - D

D - B - A - C - B - E - D - C

We cannot start from any other city.

VIEW SOLUTION

12. **A**
Consider the following scenario:

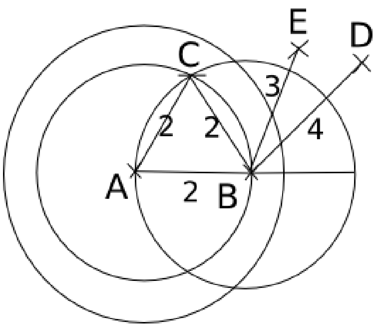


So, if a ration shop is setup on the line joining A and E, just inside the 3 km radius circle, it will be within 3 km of each city.

VIEW SOLUTION

13. **A**

Consider the following scenario:



One ration shop can be set up at C, so that it covers A,C and B. Another ration shop can be set up between E and D(such that distance of the shop from either of them is not greater than 2) so that it services both E and D.

So, two ration shops are required to be set up for them to be within 2 km of each city. Option a) is the correct answer.

VIEW SOLUTION

14. **C**

From each hub, there will be flights to 7 cities. So total total number of flights originating or terminating at each hub = 7*4 = 28. For all three hubs, it would be 28*3 = 84

There are three hubs in total. Each hub must also be interconnected. The total number of flights between any two hubs will be 4. For three hubs it will be 12.

Hence, the required number will be 84 + 12 = 96.

VIDEO SOLUTION

15. **40**

In G1, we have three cities namely A, B, C. Person living in any of these three cities should be able to travel to other city once in the morning nad one in the night. Therefore, a total of 4 flights are required between a pair of cities.

A ---> B (Morning flight)

A ---> B (Evening flight)

B ---> A (Morning flight)

B ---> A (Evening flight)

Number of flights between the cities of G1 = 3c2*4 = 12
Between cities in A and any city in G2 = 3*4 = 12
Between B and any city in G3 = 2*4 = 8
Between C and any city in G4 = 2*4 = 8
Total = 12*2 + 8*2 = 40

VIDEO SOLUTION

16. **4**

The cities those were a part of G2 will be shifted to either G3 or G4 but that will not have any impact on the number of the total flights from G1. The only reduction which will take place due to the number of flights shutting down from A to C.
Hence, the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose = 4

Alternate method :

Let us determine the number of flights under new conditions.
Flights between A and B = 4
Between B and any city in G3 = 4*4 = 16
Between C and B = 4
Between C and any city in G4 = 4*4 = 12
Hence, total flights = 36
Thus, reduction = 40 - 36 = 4.

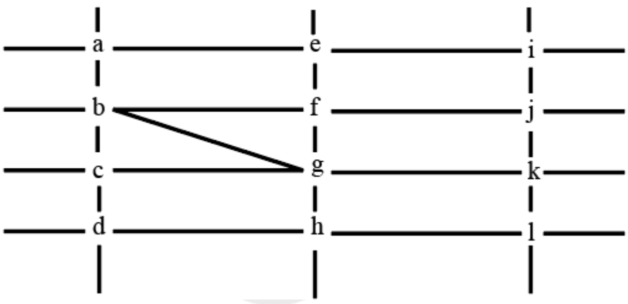
VIDEO SOLUTION

17. **C**

There are ten cities. We need to find the minimum number of flights required to travel from any city to any city. Any two cities can be selected in 10C2 ways. Now for these two cities, a person will need minimum 4 flights. (1 to go from A to B, 1 to go from B to A. Similarly, 1 to return to A and 1 to return to B) Thus, minimum number of required flights = 45*4 = 180.

VIDEO SOLUTION

18. **C**



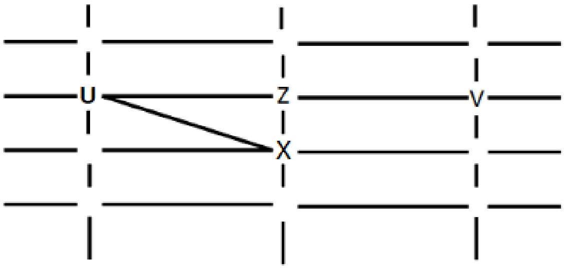
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

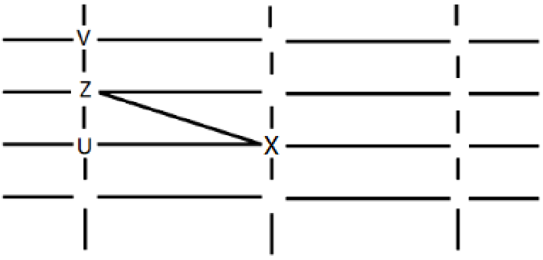
Only cases possible are:

1.



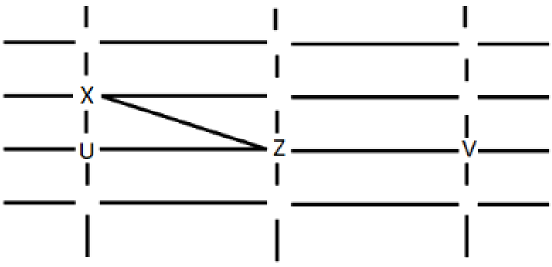
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



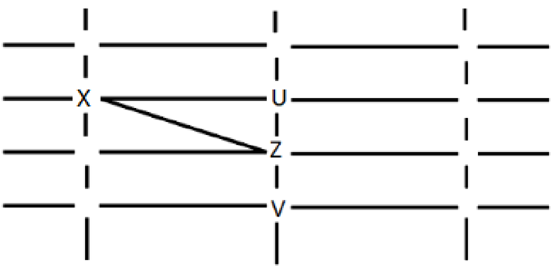
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



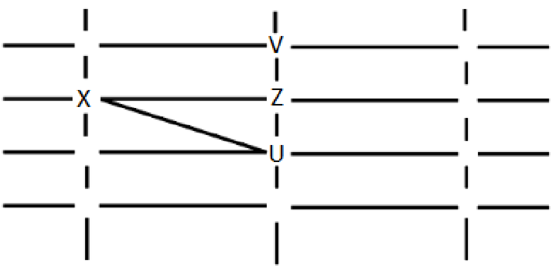
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

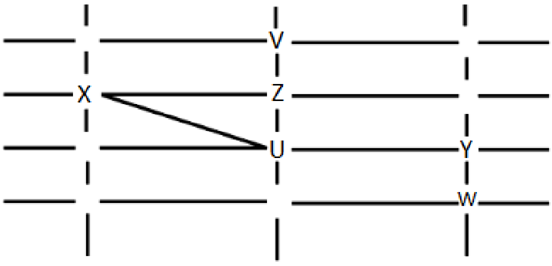


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



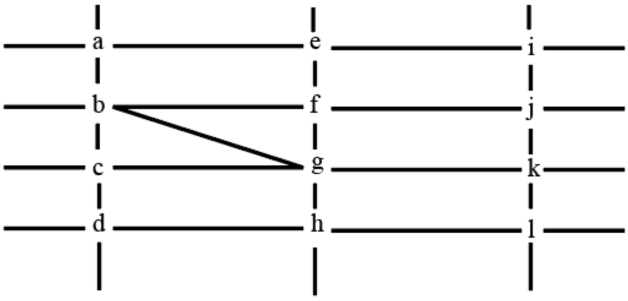
W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



V can see U and Z only. Hence C is the answer.

VIDEO SOLUTION

19. **C**



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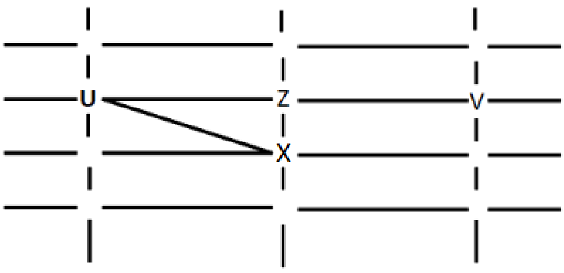
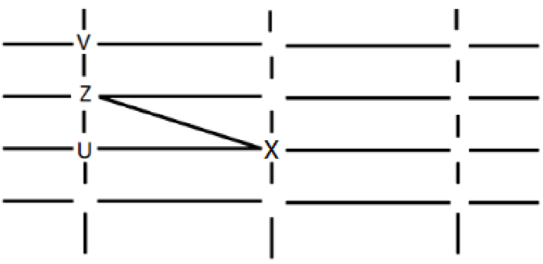
Only cases possible are:

1.



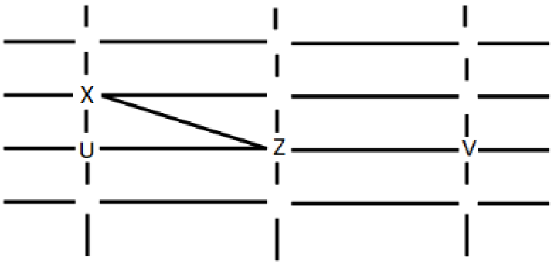
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



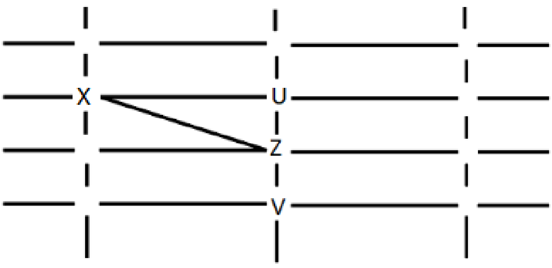
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



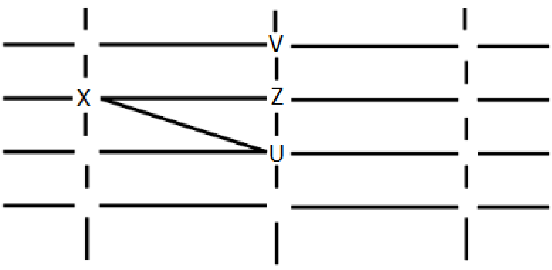
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

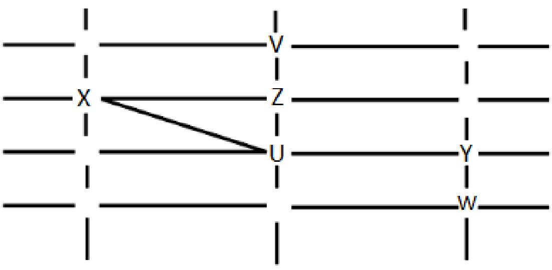


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



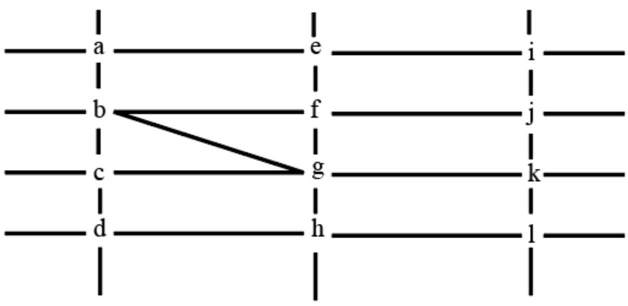
W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



To reach Y, X has to go from b to g and g to k, i.e. 2 streets.

VIDEO SOLUTION

20. **A**



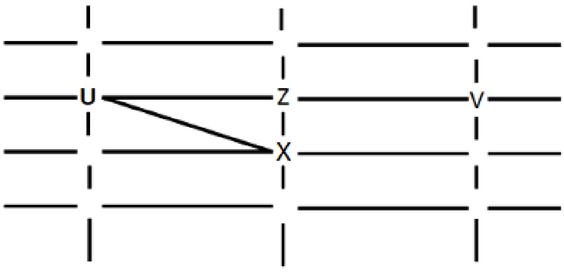
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

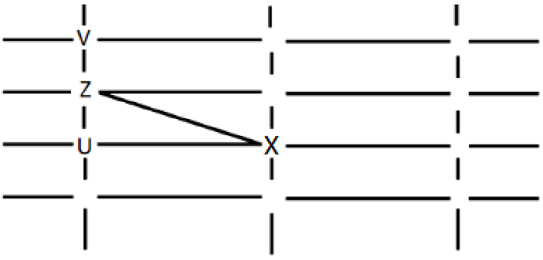
Only cases possible are:

1.



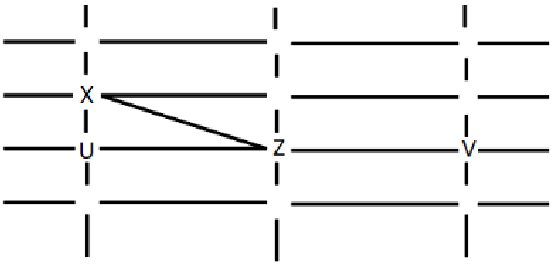
W cannot see V or Z. So W can only be at the intersection a. Since Y can see only U and W, Y can only be at c where X can see him. Hence this case is rejected.

2.



Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



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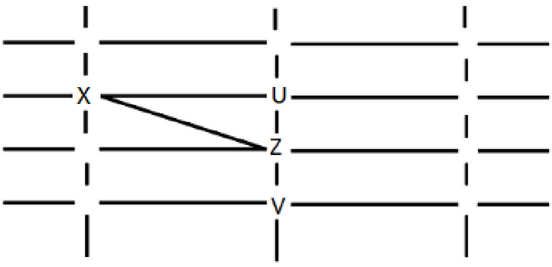
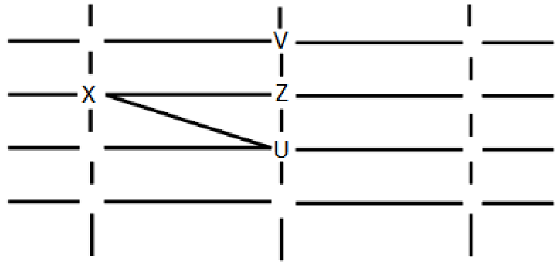
4.

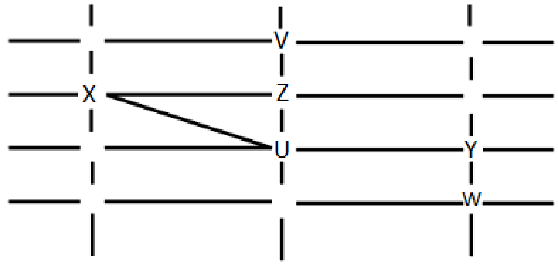


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



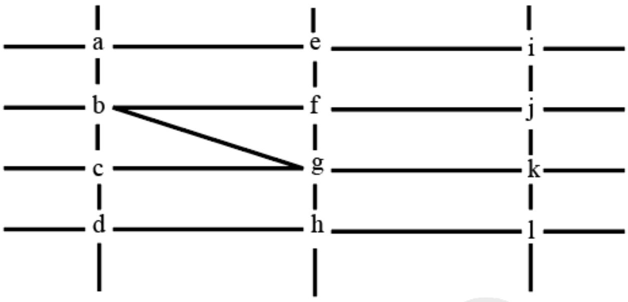
W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



If a new person stands at d(left down corner), they can see W and X only. Hence A is the answer.

VIDEO SOLUTION

21. **C**



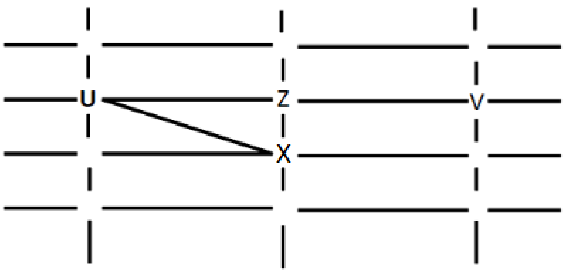
From 1, X, U, and Z are standing at the three corners of a triangle formed by three street segments.

From 2, X can see only U and Z.

From 4, U sees V standing in the next intersection behind Z. Also, no one among the six is standing at intersection d.

Only cases possible are:

1.



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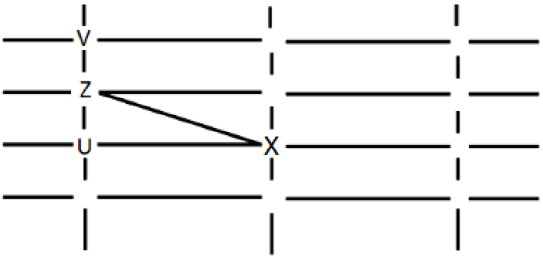
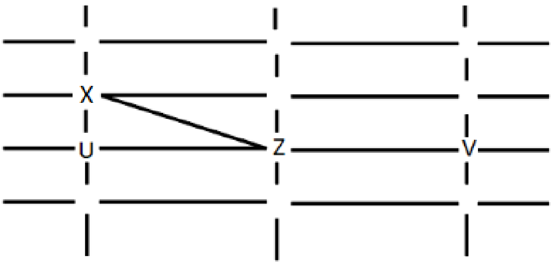
2.



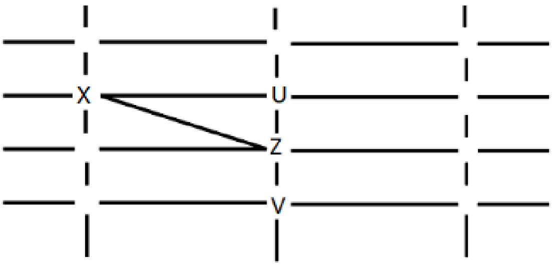
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

3.



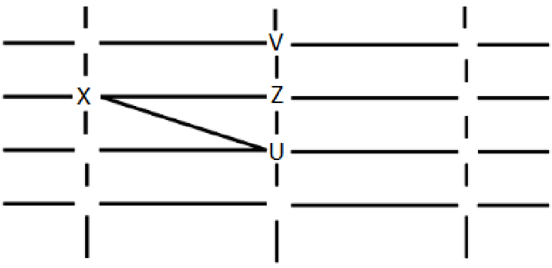
Y can only see U and W. Y cannot be placed anywhere. Hence this case is also rejected.

4.

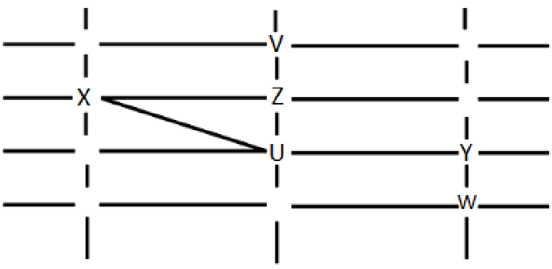


W cannot see V or Z. W can only be placed at i. Y can see only U and W. Y can only be placed at j or e, where he can see more people than U and W. Hence this case is also rejected.

5.



W cannot see V or Z. Y can only see U and W. Hence W and Y can only be placed as shown:



No one is standing at the intersection A. Hence C is the answer.

VIDEO SOLUTION

22. **D**

Option (a): When queen is at f8. In this case h8 and b4 will be under attack.

	a	b	c	d	e	f	g	h
8						(f8)		h8
7				d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1							

Option (b): When queen is at a7. In this case a3 and d7 will be under attack.

	a	b	c	d	e	f	g	h
8								h8
7	(a7)			d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1							

Option (c): When queen is at c1. In this case a1 and a3 will be under attack.

	a	b	c	d	e	f	g	h
8								h8
7				d7				h7
6								
5								
4		b4						
3	a3							
2								
1	a1		(c1)					

Option (d): When queen is at d3. In this case a3, d7 and h3 will be under attack.

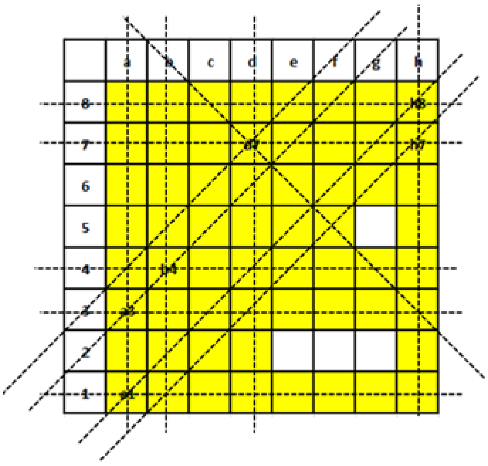
	a	b	c	d	e	f	g	h
8								h8
7				d7				h7
6								
5								
4		b4						
3	a3			(d3)				
2								
1	a1							

Therefore, we can say that option D is the correct answer.

VIDEO SOLUTION

23. **C**

From the diagram we can see that except positions e2, f2, g2 and g5 queen can attack at least one among the given pieces.



Hence, we can say that there are exactly for position from where queen can't attack any of the given pieces. Therefore, option C is the correct answer.

VIDEO SOLUTION

24. **C**

Let us draw the diagram and mark position of various pieces as given in the question.

	a	b	c	d	e	f	g	h
8								
7								
6								
5			Queen				g5	
4								
3	a3						g3	
2			c2					
1							g1	

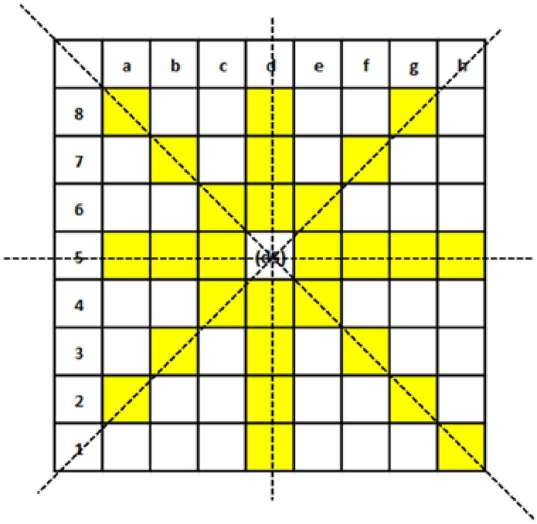
Attack line is shown by the yellow color. All the pieces on this line will be under attack.

	a	b	c	d	e	f	g	h
8								
7								
6								
5			Queen				g5	
4								
3	a3						g3	
2			c2					
1							g1	

From the diagram we can see that a3, g1, c2 and g5 are under attack. Hence, option C is the correct answer.

VIDEO SOLUTION

25. **C**



From the diagram we can see that the number of positions those are safe from queen's attack = 6 + 5 + 5 + 5 + 5 + 5 + 5 = 36. Therefore, option C is the correct answer.

VIDEO SOLUTION

26. **A**

Let us consider a 5x5 matrix. Let us start with the top left square and fill number 1 in as many squares as possible.

1		1		1
1		1		1
1		1		1

We have to use a second number, 2 to fill the gap between two 1s.

1	2	1	2	1
1	2	1	2	1
1	2	1	2	1

All the cells in row 2 and row 4 are adjacent to the cells containing numbers 1 and 2. Therefore, rows 2 and 4 should be filled with a new set of numbers. We need at least 2 numbers to fill a row such that the adjacent cells do not contain the same number (by alternating the numbers in the consecutive cells). Rows 2 and 4 are completely isolated from each other and hence, the same set of numbers can be used to fill both the rows.

1	2	1	2	1
3	4	3	4	3
1	2	1	2	1
3	4	3	4	3
1	2	1	2	1

4 numbers are required to fill a 5x5 matrix.
It has been given that we are allowed to make 1 mistake - One pair of adjacent cells can contain the same number. In the arrangement given above, we can alter any value along the edge to satisfy this condition. For example, the 2 in the bottom-most row can be changed to 4. Still, the number of numbers required to fill the matrix will be 4.

Another way to approach this problem is as follows:
We know that a minimum of 4 numbers are required to fill a 5x5 matrix. If we are allowed to make a mistake, then the number of numbers required should either remain the same or go down. 4 is the smallest value among the given options. Therefore, we can be sure that even if we are allowed to make a mistake, 4 numbers will be required to fill the matrix and hence, option A is the right answer.

VIDEO SOLUTION

27. **D**
It has been given that all the cells adjacent to a cell must have different numerals. Let us start filling the matrix from the central square since the central square has the maximum number of squares adjacent to it (8) and it will be easier to work around the central 9 squares. A minimum of 9 numbers will be required to fill the central 9 squares.

	2	3	4	
	9	1	5	
	8	7	6	

Now we have to fill the remaining squares. Let us start with the top left square. We have to check whether the 9 numbers will be sufficient to fill all the squares such that no 2 squares adjacent to a square have the same number. We can use any of the 3 numbers 4, 5, and 6 to fill the top left square since none of the numbers in the second column are adjacent to these numbers.

Let us assume that we use 4 to fill the top left square. Now, one of the cells with the number 4 has become adjacent to the cell with number 2 and no other cell adjacent to cell with number 2 (in the second row and second column) can have 4 as its neighbour. Similarly, we can fill the first row with numbers 8 and 7.

4	7	8		
5	2	3	4	
6	9	1	5	
	8	7	6	

In essence, we are trying to create a gird around each of the numbers in the corners of the inner 3x3 matrix such that no 2 cells adjacent to a cell have the same number. Filling the other cells similarly, we get the following matrix as one of the possible cases.

4	7	8	6	7
5	2	3	4	9
6	9	1	5	2
3	8	7	6	8
2	5	4	3	9

We need a minimum of 9 numbers to fill a 5x5 matrix such that for any cell, no 2 cells adjacent to it contain the same value. Therefore, option D is the right answer.

VIDEO SOLUTION

28. **4**

Let us consider a 5x5 matrix. Let us start with the top left square and fill number 1 in as many squares as possible.

1		1		1
1		1		1
1		1		1

We have to use a second number, 2 to fill the gap between two 1s.

1	2	1	2	1
1	2	1	2	1
1	2	1	2	1

All the cells in row 2 and row 4 are adjacent to the cells containing numbers 1 and 2. Therefore, rows 2 and 4 should be filled with a new set of numbers. We need at least 2 numbers to fill a row such that the adjacent cells do not contain the same number (by alternating the numbers in the consecutive cells). Rows 2 and 4 are completely isolated from each other and hence, the same set of numbers can be used to fill both the rows.

1	2	1	2	1
3	4	3	4	3
1	2	1	2	1
3	4	3	4	3
1	2	1	2	1

As we can see, a minimum of 4 numbers are required to fill a 5x5 matrix. Therefore, 4 is the correct answer.

VIDEO SOLUTION

29. **4**

Let us use 1 to denote the first number that we fill. We have to fill as many squares with 1 as possible. If we start with the top-left square, we can fill 4 squares with the number 1.

1		1
1		1

Now, we can fill number 2 only in 2 of the 5 squares available.

1	2	1
1	2	1

The 3 squares available now are adjacent to each other. Therefore, we will require at least 2 numbers to fill these squares.

1	2	1
3	4	3
1	2	1

We need a minimum of 4 numbers to fill a 3x3 square matrix such that no 2 adjacent cells contain the same number. Therefore, 4 is the correct answer.

VIDEO SOLUTION

30. **D**

We know that quantity between Vaishali and jyotishmati is 300,

So quantity in avanti-vaishal route should be 400+300=700.

So free capacity is 1000-700=300. Hence option D .

VIDEO SOLUTION

31. **D**

We know that quantity between Vaishali and jyotishmati is 300, So quantity in avanti-vaishal route should be 700. Now at jyotishmati the required quantity is 400+700 = 1100.

But through vaishali only 300 comes , so 800 should come through Vidisha-jyotishmati route.

Now demand at vidisha is 200. So total quantity required in avanti - vidisha route is 800+200=1000.

Hence, free capacity available = 1000-1000=0. Hence option D.

VIDEO SOLUTION

32. **D**

We know that quantity between Vaishali and jyotishmati is 300,

So quantity in avanti-vaishal route should be 700.

Now at jyotishmati the required quantity is 400+700 = 1100.

But through vaishali only 300 comes , so 800 should come through Vidisha-jyotishmati route.

Now demand at vidisha is 200. So total quantity required in avanti - vidisha route is 800+200=1000. Hence option D.

VIDEO SOLUTION

33. **48**

Travel time between A and B = $(10 \times 3) + (7 \times 1) + (2 \times 2) = 41$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required for the north-south lines = $\left(\frac{60}{15}\right) \times 2 \times 2 = 16$

Travel time between M and N = $(20 \times 2)+(17 \times 1)+(2 \times 2)=61$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from M to N, then the latest by which that train will start from N to M will be at 7:20 am, as in the east-west direction, a train starts from M and N every 15 minutes.

So the total no. of trains required for the east-west lines= $\left(\frac{80}{10}\right) \times 2 \times 2=32$

Total no. of trains required to service the city = 16+32 = 48

VIDEO SOLUTION

34. **8**

Travel time between A and B = $(10 \times 3)+(7 \times 1)+(2 \times 2)=41$ minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required = $\left(\frac{60}{15}\right) \times 2=8$

VIDEO SOLUTION

35. **A**

In the east-west direction, a train starts from station M every 10 minutes.

Now the first train leaving station M is at 6:00 am.

Since every 10 minutes trains are available in east-west direction, the train timings leaving from M will be like 6:10 am, 6:20 am, 6:30 am and so on.

Given, Hari has reached station M by 8:05 am.

So the earliest by which Hari can catch a train from station M is 8:10 am.

Now there are 19 stations between M and n, out of which two stations are junctions.

Time taken to travel between two stations in the east-west direction is 2 minutes.

Therefore, the time for which the train was running between M and N (excluding the stoppage time) = $20 \times 2=40$ minutes

Stoppage time at a junction is 2 minutes, while at the rest of the stations, it is 1 minute each.

Stoppage time for the train running between M and N = $(17 \times 1)+(2 \times 2)=\ 21$ minutes

Therefore, total travel time = 40+21 = 61 minutes.

So the time by which Hari reaches N is 8:10 am + 61 minutes = 9:11 am

VIDEO SOLUTION

36. **A**

Priya can reach S from T via R or V.

Case 1:- T-V-S

In the east-west direction, the first train from P arrives at T at time = 6 am + $(4 \times 2)+(3 \times 1)=\ 11$ minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Priya boards a east-west train then Priya will board a train for V from T at 10:33 am.

There are 9 stations between T and V

Travelling time between T and V = $(10 \times 2)+(9 \times 1)=29$ minutes

Therefore, Priya will reach V latest by 10:33 am + 29 minutes = 11:02 am

In the north-south direction, the first train from D arrives at V at time = 6 am + $(3 \times 3)+(2 \times 1)= 11$ minutes = 6:11 am

Since V is a junction so this train will halt for 2 minutes at V and leave at 6:13.

Since every 15 minutes, a train starts from D in the north-south direction, so the latest by which Priya will be able to board such a train from V is at 11:13 am.

There are 3 stations between V and S

Travelling time between R and S =$(4 \times 3)+((3 \times 1))= 15$ minutes

Time by which she reaches S = 11:13 +15 minutes = 11:28 am

Case 2:- T-R-S

In the north-south direction, the first train from B arrives at T at time = 6 am + $(3 \times 3)+(2 \times 1)= 11$ minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Since every 15 minutes a train starts from P in the east-west direction so the latest by which Priya will be able to board such a train is at 10:28 am.

Now since she will be able to board a north-south train earlier than the east-west train so Priya will board a train for R from T at 10:28 am.

There are 3 stations between T and R

Travelling time between T and R = $(4 \times 3)+((3 \times 1))= 15$ minutes

Therefore, Priya will reach R latest by 10:43 am

In the east-west direction, the first train from M arrives at R at time = 6 am + $(4 \times 2)+(3 \times 1)= 11$ minutes = 6:11 am

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Since every 10 minutes, a train starts from M in the east-west direction, so the latest by which Priya will be able to board such a train is at 10:43 am.

There are 9 stations between R and S

Travelling time between R and S = $(10 \times 2)+(9 \times 1)=29$ minutes

Time by which she reaches S = 10:43 +29 minutes = 11:12 am

We are getting shorter time in case 2. So 11:12 am is the answer

VIDEO SOLUTION

37. **A**

Travelling time between S and R = $(10 \times 2)+(9 \times 1)=29$ minutes

There is a stoppage of 2 minutes at R

Travelling time between R and B = $(7 \times 3)+(1 \times 2)+(5 \times 1)=28$ minutes

In the north-south direction, the first train from A arrives at R at time = 6 am +$(3 \times 3)+(2 \times 1)$ = 6:11 am.

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Every 15 minutes, a train starts from A in the north-south direction.

The last train that leaves A will be at 12:00 am and it will leave R at 12:13 am, so Haripriya must reach R till 12:13 am.

Travelling time between S and R = $(10 \times 2) + (9 \times 1) = 29$ minutes

So Haripriya must board the train at S by 11:44 pm

In the east-west direction, the first train from N arrives at S at time = 6 am +$(6 \times 2) + (5 \times 1)$ = 6:17 am.

Since S is a junction so this train will halt for 2 minutes at S and leave at 6:19.

Every 10 minutes, a train starts from N in the east-west direction.

Therefore, Haripriya should board the train which leaves S at 11:39.

VIDEO SOLUTION

38. **35**
Points to be noted:
1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse(Prefers minimum distance).
In the question, it is given that last location is A. The demand in the remaining places should be greater than A.
This implies A demand cannot be 70 units. Therefore, it is 50 units.
The demand of the location D is 30 or 50 units. This implies this should be placed before D.
The demand of the location placed before D should be greater than or equal to 50 units. Location supplier visited before D is B(60 units of demand). It cannot be C because values of C is greater than the values of B.
Therefore, order is C - B - D - A.
From warehouse to C - 12 km
C to B - 4 km
B to D - 12 km
D to A - 7 km (through warehouse)
Total distance covered = 12 + 4 + 12 + 7 = 35 km

This question is removed from the paper because if the order is CBDA, the supplier will go to A from B (and hence the last city visited will be D).

VIDEO SOLUTION

39. **38**
Points to be noted:
1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse(supplier prefers minimum distance).

In the question, it is given that total number of units delivered is 250 units.

Maximum number of widgets that can be delivered is 70 units(A) + 50 units(D) + 60 units(B) + 100 units(C) = 280 units

From this 30 units should be decreased. 30 units can be decreased only when C's demand decreases to 70 units(because the difference for remaining locations is 20 units)

Therefore, the only possibility is 70 units(A) + 50 units(D) + 60 units(B) + 70 units(C)

From statement 1, the order should be A - C - B - D(as A is nearer to warehouse than C)

Distance from Warehouse to A is 5 km

Distance from A to C is 17 km

Distance from C to B is 4 km

Distance from B to D is 12 km

Total distance covered = 5 + 17 + 4 + 12 = 38 km

40. **A**
It is given that the first location visited from the warehouse is A.

If A's demand is 50 units, C's demand should be less than 50 units which is not possible. This implies demand of locations A and C is 70 units.

A(70 units) -> C(70 units)

Warehouse to A - 5km

A to C - 17 km

Distance covered = 5 + 17 = 22 km

Remaining distance = 40 - 22 = 18 km

C to B - 4 km

B to D - 12 km

Distance covered = 4 + 12 = 16 km $\neq$ 18 km

C to D - 6 km

D to B - 12 km

Distance covered = 6 + 12 = 18 km

Therefore, supplier can cover distance 18 km if he visits D before B, i.e. demand of D should be more than demand of B. This is only possible when D's demand is 50 units and B's demand is 40 units.

It is given,

D - 50 units - 60% probability

B - 40 units - 30% probability

Required value = 0.6*0.3 = 0.18 = 18%

The answer is option A.

41. **A**
Points to be noted:
1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse(Prefers minimum distance).
Demand in all other locations should be less than or equal to the demand in first location. In the question, it is given that A is not first location. B and D cannot be first location. This implies C should be the first location.
It is given, total route distance is 29 km.
Warehouse to C is 12 km.
This implies remaining distance should be 29-12, i.e. 17km

This is only possible when C visits B, A and D later(4 + 6 + 7).
The order should be C - B - A - D
C's demand can be 70/100, B's demand should be 60, A's demand should be 50 and D's demand can be 30/50.
The possible number of widgets delivered can be
70 + 60 + 50 + 30 = 210
70 + 60 + 50 + 50 = 230
100 + 60 + 50 + 30 = 240
100 + 60 + 50 + 50 = 260
The answer is option A.

VIDEO SOLUTION

42. **D**

Maximum number of widgets delivered in a day is 100 units(C) + 70 units(A) + 60 units(B) + 50 units(D), i.e. 280 units

Given, total number of widgets delivered is 260 units. This implies 20 units must be decreased from any one of the locations.

A - (70,50), B - (60,40), C - (100,70) and D - (50,30)

20 units can be decreased from A, B or D.

Demand at location C will be 100 units and supplier first visits C.

In the question, it is also given that the route ends at B.

C(100 units), _, _, B

If B's demand is 60 units, D's demand should be more than 60 units which is not possible. Therefore, B's demand should be 40 units.

20 units is decreased at location B. This implies demand at location A is 70 units and at location D is 50 units.

Order will be C(100 units) - A(70 units) - D(50 units) - B(40 units).

It is given,

C - 100 units - 70%

A - 70 units - 60%

D - 50 units - 60%

B - 40 units - 30%

Required value = 0.7*0.6*0.6*0.3 = 0.0756 = 7.56%

The answer is option D.

VIDEO SOLUTION

43. **A**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	 B	 D	 D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	 B	 C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, $x \neq 6$.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Column 1,2,3,4 have 36, 52, 49, 68 trees respectively.

Hence A is correct answer.

VIDEO SOLUTION

44. **B**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	B	D	D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	B	C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, x$\neq$ 6.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

∴ Number of trees per plot is least for Benna=3.

VIDEO SOLUTION

45. **A**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	B	D	D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	B	C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, $x \neq 6$.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Number of PIne trees received by Chitra = 18.

VIDEO SOLUTION

46. **A**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	B	D	D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	B	C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, $x \neq 6$.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Sequence of trees received by Abha, Bina, Chitra and Dipti is 50,69,30,56.

VIDEO SOLUTION

47. **B**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	B	D	D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	B	C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, $x \neq 6$.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

Bina got 28 pine trees, Option B is correct answer.

VIDEO SOLUTION

48. **C**

There are 12 plots and each of them got even number of plots. So, possible cases are 4,4,2,2 or 6,2,2,2.

From 4, A and B got more plots than D. So, the only possible case is A, B each got 4 and C,D each got 2.

From 6, D has to get two adjacent plots and From 8, plots of C, D are nit adjacent to each other => D must have got plots in X3, X4.

C already has two plots in X1, Z2. So, the corner plot Z4 should belong to B.

From 7, B has a plot in each row and each column. So, X2 should belong to B.

Now, out of the remaining Y2, Y3, Y4 and Z3 three plots belong to A and one belongs to B.

Till now B hasn't got any plot in Third column and 2nd row.

So, Y3 belongs to B and Y2, Y4, Z3 belongs to A.

Let the number of trees in Y4 be 4x from 3, number of trees in Y3, Y2 will be 2x, x respectively.

The number of teak trees=7x+21

∴ Number of mango trees=14x+42

The table now looks like:

	1	2	3	4	
X	12 C	 B	 D	 D	14x+42
Y	21 A	x A	2x B	4x A	7x+21
Z	 B	 C	9 A	28 B	

Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees and from 2, B didn't have the largest number of trees in a plot => x<8.

x can't be 7,5,3,2,1 as for these cases at least one of x,2x,4x is neither multiple of 3 or 4.

x can be 6 or 4.

If x=6, number of Teak trees will be 63 and Mango trees will be 126 => Number of Pine trees= 205-126-63=16 but number of trees in Z3+Z4>16 so, $x \neq 6$.

If x=4, Number of Teak trees=49 and Mango trees=98 => Number of Pine trees=58. Valid case.

Number of trees with A= 30+5x=50.

From 1, number of trees with C, D= 30, 56 respectively.

So, number of trees in Z2= 18.

∴ Number of trees with B= 205-50-30-56=69.

From 2, largest number of trees in a plot is 32. They can be in the plot of either B or D. If they are from B, they have to be from X2 but in that case number of trees in Z1=1 which is neither a multiple of 3 or 4.

So, highest number of trees in a plot are with D and it is 32 -=> number of trees in X3, X4 are 32, 24 in any order.

So, number of trees in X2= 98-56-12=30

∴ Number of trees in Z1=69-30-28-8=3.

The final table will look like:

	1	2	3	4	
X	12 C	30 B	32/24 D	24/32 D	98
Y	21 A	4 A	8 B	16 A	49
Z	3 B	18 C	9 A	28 B	58

∴ Number of Mango trees=98.

VIDEO SOLUTION

49. **B**

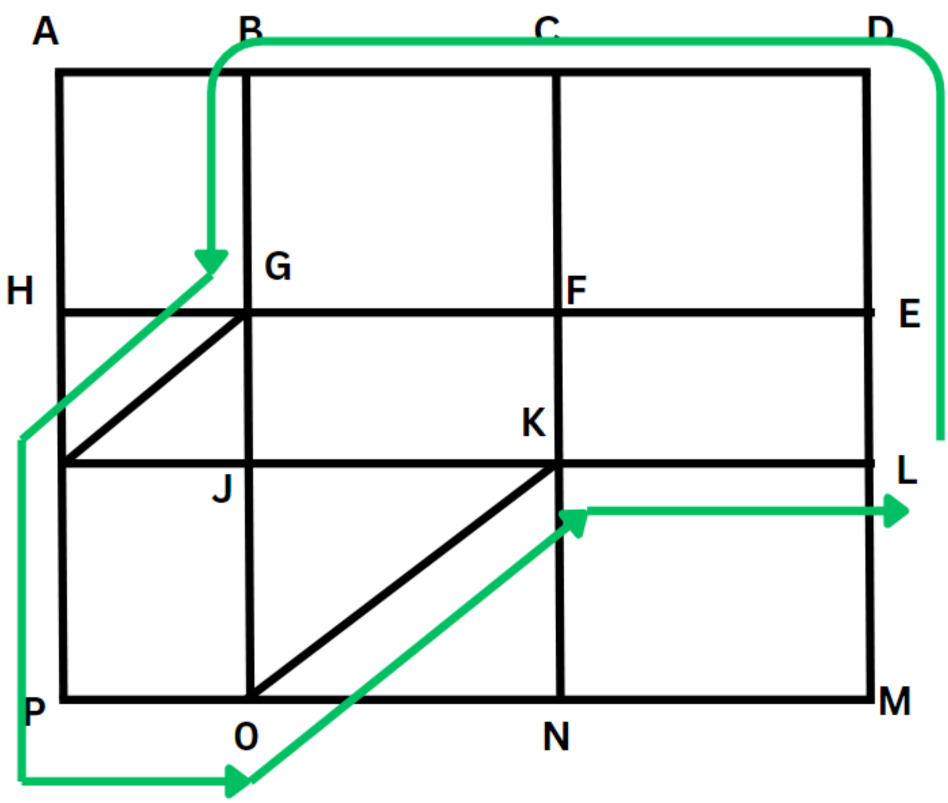
The first thing to realise here is the lengths of the paths.

KN should be equal to LM, giving the length of KO using Pythagoras theorem as 500m

Similarly, the length of HJ=OP=150m and length of GJ=EL=200m, giving the length of HG as 250 m

The shortest path from L to B and then to P (or the other way around would involve) using these hypotenuses as much as possible instead of the two adjacent sides. The shortest can be visualised as shown below or multitude of others variations, as there are multiple ways that would make one travel the shortest distance)

The below figure is the simplest one for visualisation.



Other possible paths are L-E-F-C-.. and following the same path.

The shortest distance in each of these instance would be LE+ED+DC+CB+BG+GI+IP+PO+OK+KL

Which would be 200+400+300+300+400+250+400+150+500+300 = 3200

Therefore, Option B is the correct answer.

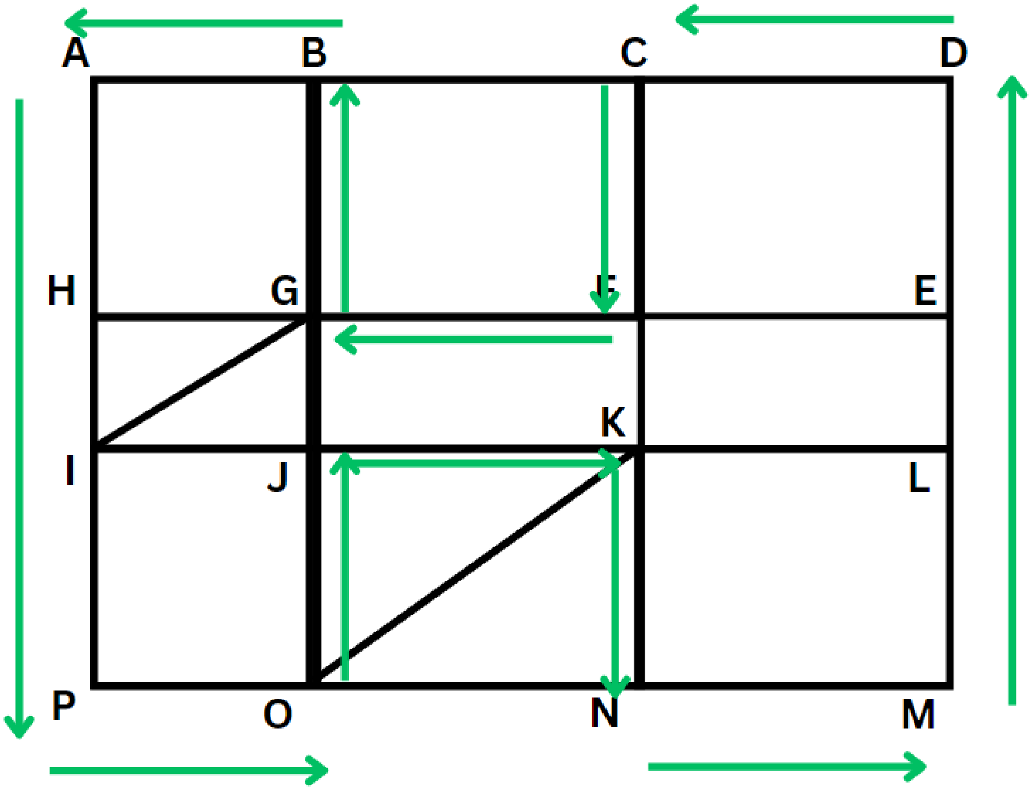
VIDEO SOLUTION

50. **5100**

Counter to the first question, we should minimize the use of those hypotenuse walkways as they reduce the distance we travel.

The longest possible route would include then travelling through the edges of the rectangles and sqaures, and not touch any point more than once.
Best case scenario would be us touching each point exactly once.

This can be visualized as:



Where we do not use any of the hypotenuse and cross through each gate exactly once.

The total distance would be:

AH+HI+ IP = 400+200+400 = 1000

DE+EL+LM = 400+200+400 = 1000

PO+JK+NM = 150+300+300 = 750

DC+FG+BA = 150+300+300 = 750

JO+KN = 400+400 = 800

CF+GB = 400+400 = 800

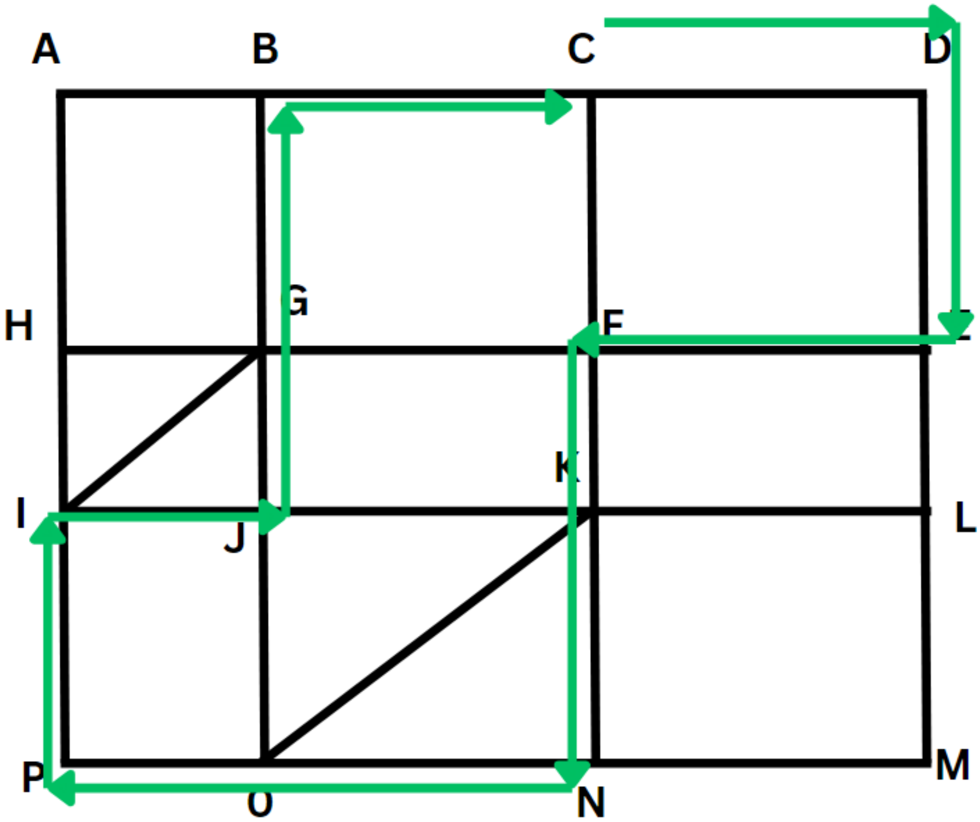
Adding up to: 5100m

Therefore, 5100 is the correct answer.

VIDEO SOLUTION

51. **3500**

Similar to the previous question, we should try to avoid the hypotenuse



The total path distance would be 300(CD) + 400(DE) + 300(EF) + 200(FK) + 400(KN) + 300(NO) + 150(OP) + 400(PI) + 150(IJ) + 200(JG) + 400(GB) + 300(BC)

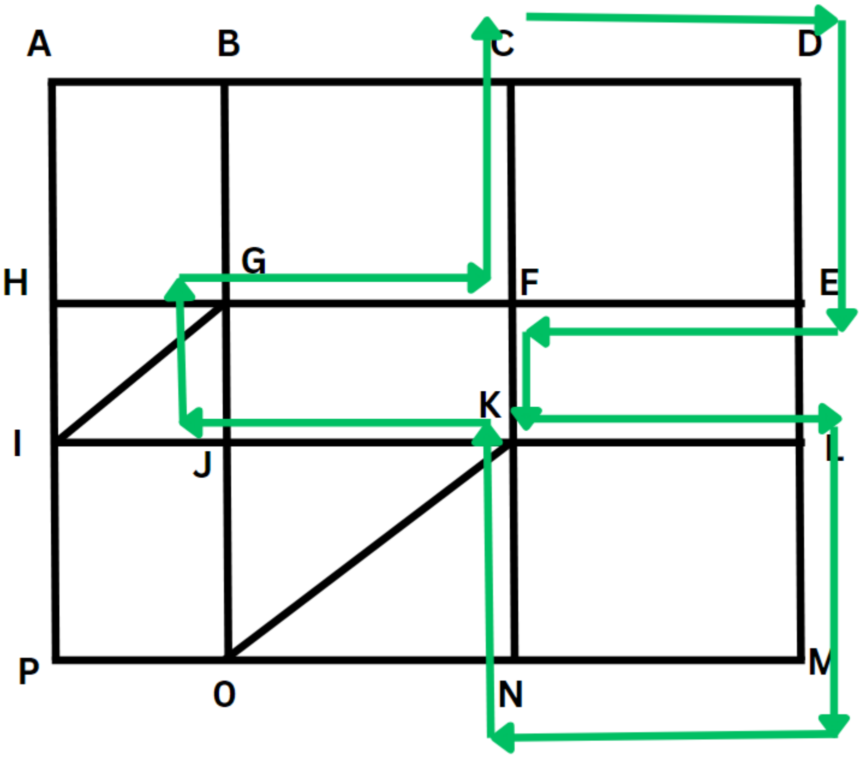
Adding up to 3,500 meters

Therefore, 3500 is the correct answer.

VIDEO SOLUTION

52. **C**

Since we can only walk along the side of lakes, that drastically reduces the paths we can take.
The diagram below shows the path with the maximum distance travelled.



The path is CD-DE-EF-FK-KL-LM-MN-NK-KJ-JG-GF-FC

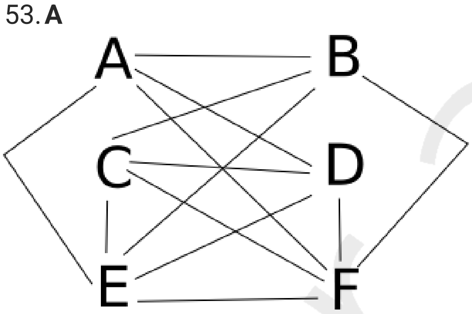
(the reverse of this path is also valid)

We can either manually add the lengths or use shorter methods to note that in the path we travel walkways of length 400 m 4 times (DE, LM, KN, EF), walkways of length 300m 6 times (CD, EF, KL, MN, KJ, GF) and walkways of length 200 m 2 times (FK, JG)
Giving the total length to be 1600+1800+400 = 3800

Therefore, Option C is the correct answer.

VIDEO SOLUTION

53. **A**



Consider the following diagram: In this, the nodes E and F are connected to all the other nodes whereas the other four nodes are connected to four nodes each.

The total number of connections is 13.

VIEW SOLUTION

54. **B**
4 first level nodes are connected to A => 4 ways.

Each of these first level nodes are connected to 1 second level node by four connections => 4 ways

Each of these second level nodes are connected to 2 third level nodes => 2 ways.

Each of these third level nodes are connected to B by 1 route => 1 way

=> 4 * 4 * 2 * 1 = 32 ways.

VIEW SOLUTION